

ESSENTIAL MATHEMATICS GRADE 10: VOLUME AND CAPACITY

CBE Phenomenon-Driven Lesson Sequence — Sub-Strand 2.7 (8 Lessons)

SUB-STRAND OVERVIEW	
Grade Level	10
Subject	Essential Mathematics
Strand	Strand 2.0: Measurements and Geometry
Sub-Strand	Sub-Strand 2.7: Volume and Capacity
Total Duration	8 lessons × 40 minutes = 320 minutes total
Sub-Strand Content	– Volume and capacity of cones: deriving and applying the formula $V = \frac{1}{3}\pi r^2 h$; relating to the volume of a cylinder of equal base and height – Volume and capacity of pyramids (square and rectangular base): deriving and applying $V = \frac{1}{3} \times \text{base area} \times \text{height}$ – Volume and capacity of frustums of cones: applying $V = \frac{1}{3}\pi h(R^2 + Rr + r^2)$ using dimensions of the original cone and the removed top cone – Volume and capacity of frustums of pyramids: applying $V = \frac{1}{3}h(A_1 + A_2 + \sqrt{A_1 A_2})$ for square and rectangular bases – Relationship between volume units and capacity units ($\text{cm}^3 \leftrightarrow \text{mL}$, $\text{m}^3 \leftrightarrow \text{L}$; $1 \text{ L} = 1\,000 \text{ cm}^3$) – Reading and interpreting measurements of real containers (buckets, grain stores, water tanks, cooking pots) shaped as cones, pyramids and frustums – Selecting and applying appropriate formulae to solve real-life volume and capacity problems – Project: making and measuring model containers shaped as frustums, pyramids and cones from locally available materials
Learning Outcomes	By the end of the sub-strand, the learner should be able to: - determine the volume and capacity of cones and pyramids, - compute volume and capacity of a frustum of a cone, - calculate the volume and capacity of a frustum of a pyramid, - apply volume and capacity in real-life situations, - appreciate volumes and capacities in real life situations.
Core Competencies	– Critical Thinking and Problem Solving: the learner acquires open-mindedness and creativity skills when discussing with peers and working out the volume and capacity of containers shaped as cones, pyramids and frustums; learners must choose and justify the correct formula for each shape encountered. – Creativity and Imagination: the learner develops connection and networking skills when deriving the volume of frustums by subtracting a smaller cone or pyramid from a larger one, and when constructing physical models of these solids from locally available materials such as cardboard, old packaging and clay.
Core Values	– Unity: the learner displays tolerance of different opinions while working with peers when constructing models of frustums as learning resources, sharing materials equitably and dividing tasks fairly within the group. – Respect: the learner listens to others' opinions while discussing appropriate formulae for working out volume and capacity of containers, and acknowledges contributions of group members during presentations.

Science & Engineering Practices	– Asking Questions and Defining Problems: learners pose questions about why different containers (buckets, grain stores, water tanks) hold different amounts even when they look similar in height, initiating the mathematical investigation of volume and capacity. – Developing and Using Models: learners build physical models of cones, pyramids and frustums to visualise the relationship between a frustum and its parent solid, then use mathematical formulae as abstract models to predict capacity. – Using Mathematics and Computational Thinking: learners select and apply the appropriate volume formula, substitute measured dimensions, and convert between volume units (cm^3, m^3) and capacity units (mL, L) to solve real-life container problems. – Constructing Explanations: learners explain why $V(\text{frustum}) = V(\text{large cone}) - V(\text{small cone})$ by referencing their physical models and measurements, justifying each step of their working.
Pertinent & Contemporary Issues (PCIs)	– Environment and Technology: the learner uses appropriate containers from the local environment — such as plastic buckets used for carrying water from the borehole, metal grain-storage bins common in Rift Valley farms, and cooking pots used for ugali — to work out the capacity of cones, pyramids and their frustums, promoting environmental awareness and resourcefulness. – Education for Sustainable Development: by calculating the capacity of water-harvesting tanks and grain-storage structures shaped as frustums, learners connect mathematics to sustainable food and water management practices relevant to Kenyan farming communities.
Career Connections	– Civil and Structural Engineering: calculating volumes of conical and pyramidal roof structures, water tanks and grain silos used in construction projects across Kenya. – Architecture and Interior Design: determining capacity of decorative containers, lamp shades and furniture components with frustum or pyramidal shapes. – Agriculture and Food Science: estimating grain storage capacity of pyramidal or frustum-shaped stores; calculating volumes of conical heaps of maize or beans at market stalls in Wakulima Market, Nairobi. – Manufacturing and Packaging: designing and sizing packaging (e.g., paper funnels, ice-cream cones, juice cartons) for the food and beverage industry in Kenyan factories. – Hospitality and Catering: working out capacities of cooking vessels and serving containers in hotels and restaurants along the Kenyan coast and urban centres such as Kisumu and Mombasa. – Water Resources Management: computing the volume of water stored in conical or frustum-shaped tanks for community water projects in arid and semi-arid areas of Kenya.
Focus for Lessons	This unit develops learners' ability to calculate the volume and capacity of cones, pyramids and their frustums, rooting every concept in Kenyan everyday objects such as water-carrying buckets, grain-storage bins and cooking pots. Learners move from establishing the cone–cylinder relationship through hands-on measurement of real containers to applying and justifying formulae for frustums of both cones and pyramids. The unit culminates in a project in which learners design, build and document model containers, connecting geometry to sustainable resource management in the Kenyan context.
Driving Question / Key Inquiry Questions	DRIVING QUESTION: Why do the tea farmers' containers — a cone, a cylinder and a bucket — look nearly the same size but hold such different amounts of water, and how can we use mathematics to design a frustum-shaped container that holds exactly the capacity we need? KICD KEY INQUIRY QUESTION: 1. Why is the knowledge of volume and capacity useful?
Anchoring Phenomenon	A group of farmers in Kericho have three different containers they use to store harvested tea leaves before collection: a tall cylindrical drum, a conical heap-cover made from woven sisal, and a metal bucket (frustum of a cone) used for carrying leaves to

	the weighing station. All three containers have roughly the same height and the same opening diameter, yet when filled with water the cylindrical drum holds exactly three times as much water as the conical cover, and the metal bucket holds noticeably more water than the cone but less than the cylinder. Learners observe these three containers side by side, filled with water, and are puzzled: why do shapes that look so similar in size hold such dramatically different amounts? This anchoring phenomenon drives the entire unit — learners must unpack the mathematical relationship between a cone and a cylinder, then extend that reasoning to pyramids and frustums, to explain and predict the capacities of the containers used every day by Kenyan tea farmers.
Storyline Thread	Lesson 1 — Predict & Anchor Phenomenon: Learners examine the three Kericho tea-farmer containers (or physical models/images thereof). They predict which holds the most water, record estimates, and articulate what they wonder. The driving question is introduced and posted on the Driving Question Board. Learners review the relationship between volume units (cm^3, m^3) and capacity units (mL, L) as prerequisite knowledge. Lesson 2 — Cone vs. Cylinder (Observe & Explain): Using a cone and cylinder of identical base radius and height (models or demonstration containers), learners fill the cone with water and pour it into the cylinder three times to observe that $V(\text{cylinder}) = 3 \times V(\text{cone})$. They derive the cone volume formula $V = \frac{1}{3}\pi r^2 h$, apply it to the Kericho conical cover, and calculate its capacity in litres. Learners update the DQB with new understanding. Lesson 3 — Volume of Pyramids (Observe & Explain): Learners use models of a square-based pyramid and a cuboid of equal base and height to establish $V(\text{pyramid}) = \frac{1}{3} \times \text{base area} \times \text{height}$. They apply the formula to pyramidal grain-store models, computing volumes and converting to capacity. They compare results with their Lesson 1 predictions and refine the DQB. Lesson 4 — Introducing Frustums of Cones (Explain & Model Building): Learners measure a real frustum-shaped bucket (top radius R, bottom radius r, height h) and the dimensions of the full cone from which it could be cut. They derive $V(\text{frustum of cone}) = V(\text{large cone}) - V(\text{small cone}) = \frac{1}{3}\pi h(R^2 + Rr + r^2)$ step by step. They calculate the bucket's capacity and compare it to the cylindrical drum, resolving the anchoring phenomenon. Lesson 5 — Volume of Frustums of Pyramids (Explain & Model Building): Learners extend the frustum concept to square and rectangular pyramidal frustums using $V = \frac{1}{3}h(A_1 + A_2 + \sqrt{A_1 A_2})$. They apply the formula to a pyramidal grain-storage bin model, compute capacity in litres and kilograms of maize (using density), and update their cumulative models on the DQB. Lesson 6 — Problem Solving with Real Containers (Apply & DQB Update): Learners work in groups on a set of Kenyan-context word problems: sizing a conical water tank for a homestead in Turkana, calculating the capacity of a frustum-shaped cooking pot used in a githeri recipe for 100 people, and determining how many frustum-shaped cups fill a cylindrical storage drum. Groups present solutions and justify formula choices; the DQB is updated with consolidated understanding. Lesson 7 — Project: Model Making & Measurement: Learners design and construct model containers shaped as a frustum of a cone, a frustum of a pyramid and a cone using locally available materials (cardboard from cereal boxes, old packaging). They measure dimensions, calculate theoretical volumes and capacities, then fill models with water or sand to verify experimentally. Discrepancies between theoretical and measured values are discussed. Photographs and video clips are taken for documentation. Lesson 8 — Model Building Culmination & Assessment: Groups present their project models, share calculated and measured capacities, and explain sources of error. The class revisits the anchoring phenomenon — the Kericho farmers' containers — and constructs a final annotated diagram showing the volume relationships between cone, cylinder, pyramid and frustum. A written class activity assesses all five specific learning outcomes. The DQB is completed and learners articulate what they now know

	versus what they predicted in Lesson 1.
Supporting Phenomena	– A mandazi vendor in Mombasa uses a conical paper bag and a frustum-shaped plastic cup to pack mandazi for customers; the vendor claims both hold the same number of mandazi, but customers dispute this — learners investigate whether the volumes are truly equal given the same slant height. – A water-harvesting project near Lake Turkana installs frustum-shaped underground cisterns; community members want to know how many 20-litre jerricans can be filled from a cistern of given top diameter, bottom diameter and depth — learners calculate the capacity and convert to jerricans. – A Nairobi school's mathematics club builds a square pyramidal display stand for their science fair; the teacher asks them to fill it with sand to find its volume experimentally, then compare the result to the formula — learners reconcile measured and calculated values.

ATTRIBUTION AND LICENSING

Curriculum Alignment: Structured in pedagogical alignment with the competence-based curriculum format and learning outcomes established by the Kenya Institute of Curriculum Development (KICD). This material is independently developed and is not officially endorsed or published by KICD.

Content & Pedagogy: Developed and adapted by the educator team at SeaVuria (registered 501(c)(3) in the United States and registered NGO in Kenya, No. 872-850A-BF11), who designed the pedagogical framework, lesson templates and exemplar lessons on which this material is based.

Production & Layout: Educational design and curriculum sequencing by SeaVuria. Technical layout, structural formatting, and digital optimization/implementation executed in partnership with ARES Education (registered 501(c)(3) in the United States and Canada (as Afretech Aid) and registered NGO in Kenya), with lesson drafting and compilation assisted by AI (Claude, Anthropic).

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LESSON 1 (80 minutes (2 x 40-minute lessons)): Predict and Anchor the Phenomenon: The Three Kericho Containers

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	To launch the anchoring phenomenon of the three Kericho tea-farmer containers, elicit learners' initial ideas and predictions about which holds the most, and revise the units of volume and capacity with conversions so that learners are ready to investigate how shape determines capacity.
Knowledge	Learners distinguish volume (space inside a solid, measured in cm^3 and m^3) from capacity (amount a container can hold, measured in mL and L). They recall the conversions $1 \text{ cm}^3 = 1 \text{ mL}$, $1000 \text{ cm}^3 = 1 \text{ L}$ and $1 \text{ m}^3 = 1000 \text{ L} = 1,000,000 \text{ cm}^3$. They recall volume of prisms from Junior School. They know that drum (A), cover (B) and bucket (C) are all about 60 cm tall with a 42 cm opening.
Skills	Predicting and justifying with reasoning (NGSS SEP: Asking Questions and Defining Problems; Developing and Using Models). Estimating and comparing quantities. Converting between cm^3 , m^3 , mL and L. Recording predictions and wonderings. Communicating reasoning with peers (mathematical process skills: reasoning, communication, collaboration).
Attitudes	Curiosity about why similar-looking containers hold different amounts. Respect: listening to others' opinions during discussion (KICD value). Unity: tolerating different predictions while working with peers (KICD value). Appreciation of volume and capacity in the lives of Kenyan farmers (SLO e: appreciate volumes and capacities in real life situations).
Key Inquiry Question	1. Why is the knowledge of volume and capacity useful?
Purpose in Storyline	Lesson 1 launches the unit. Learners meet Mama Chebet's drum (A), cover (B) and bucket (C), commit to predictions that are revisited in Lessons 2 to 4 and 8, and post the driving question on the DQB. The unit conversions revised here are needed in every later lesson. Actual capacities are NOT revealed; they emerge through Lessons 2 to 4.
Safety Notes	If water is used with physical containers, wipe spills immediately so nobody slips, and keep water away from electrical devices. Use plastic or card models rather than heavy metal buckets. Learners lift containers only in pairs. Use only clean water. Handle sharp metal edges of any real bucket with care, or use images or models instead.

B. LESSON OVERVIEW

Ms. Wanjiru opens the class with a story: Mama Chebet, a tea farmer in Kericho, owns three containers for tea leaves before collection: a cylindrical drum (A), a woven sisal conical cover (B) and a metal frustum bucket (C). All three are about 60 cm tall with a 42 cm opening, so they look nearly the same size. Her neighbour Mr. Kiprop, who carries leaves in the bucket to the weighing station, asks why the bucket is not as big as the drum. Learners individually predict the order of the containers by capacity, estimate in litres, and justify their reasoning. They then observe the containers (models or images with the same dimensions) side by side, record what they notice and wonder, and revise volume versus capacity units with conversion practice (without revealing actual capacities). In the Explain phase learners compare predictions, sort their reasoning and identify the mathematical idea they lack. The class builds the Driving Question Board with the driving question and learner questions. In the Model Building phase learners draw the initial class model: three containers with the shared height and opening, a prediction column, and a note that shape determines volume. This model will grow in each lesson.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Visually Comparing the Capacity and Volume of Three Containers (Metric Units) Source: CK-12  Search ARES for similar videos  EXERCISE: Estimate volume (milliliters and liters) Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Learners hear the Mama Chebet and Mr. Kiprop story and see a labelled diagram or image of drum (A), cover (B) and bucket (C) with the shared measurements: height about 60 cm and opening diameter 42 cm. Working silently and individually for 4 minutes, each learner writes: (1) their ranking from most to least water held, (2) an estimate in litres for each container, using familiar references such as a 20 L jerrycan or a 500 mL water bottle, and (3) a one-sentence reason. They then share with a partner and mark their prediction on a class tally chart on the board (e.g. $A > C > B$).	Say: 'Mama Chebet filled all three containers to the brim with water. They are all about 60 centimetres tall and 42 centimetres across the top. Mr. Kiprop asks: does the bucket hold as much as the drum? What do YOU think?' Do NOT reveal or hint at actual capacities or the one-third relationship. Say: 'Write your own prediction first. There is no wrong answer today; we want your honest thinking.' WAIT TIME: 15 seconds of silent thinking before writing begins. After pair-sharing, cold-call 4 learners with: 'Which container did you rank first, and what is your reason?' Record every prediction on the tally chart without evaluating. Circulate and note learners who reason about shape and learners who say 'same size so same amount'.	Predict-Commit-Justify: learners write a prediction and a reason before any instruction, creating a stake in the outcome and exposing the preconception that equal height and opening mean equal capacity. The class tally shows disagreement, which motivates the investigation.	Look for: (1) a complete written ranking and litre estimates for all three containers by every learner; (2) whether the reasoning mentions shape (narrowing, pointed) versus only height and width; (3) whether the litre estimates are plausible compared with familiar objects. Note the learners who predict all equal. Record the tally of rankings for use in Lesson 8.
Observe Phase  VIDEO: Comparing the Capacity and Volume of Three or More Containers (Metric Units) Source: CK-12  Search ARES for similar videos  EXERCISE: Estimate volume (milliliters and liters) Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	In groups of 4, learners examine the three containers (physical models, card/plastic versions with a 21 cm radius ratio, or large photographs with dimensions marked) placed side by side. They fill in an observation table: shape name, height, opening radius (21 cm, half of 42 cm), bottom radius, and what they notice. They also sketch each container's side view and compare outlines. Next, they complete a quick revision station on units: (a) the difference between volume (space inside, cm^3 , m^3) and capacity (amount held, mL, L); (b) conversions such as $2,500 \text{ cm}^3 = ? \text{ L}$, $3 \text{ L} = ? \text{ mL}$, $0.5 \text{ m}^3 = ? \text{ L}$, $4,000 \text{ mL} = ? \text{ cm}^3$; (c) the volume of a cuboid-shaped jerrycan or box (prism volume from Junior School) converted to litres.	Say: 'Look carefully. What is the same? What is different? Record only what you can see and measure.' Circulate and prompt: 'Where does the bucket get narrower? What happens to the cover as you go up?' Do NOT give formulas for cone, cylinder or frustum. At the units station say: 'Volume is the space inside a solid. Capacity is how much a container can hold. $1 \text{ cm}^3 = 1 \text{ mL}$, $1000 \text{ cm}^3 = 1 \text{ L}$, $1 \text{ m}^3 = 1000 \text{ L}$.' Ask groups: 'How many cm^3 are in a litre? How could you check that with a 10 cm cube?' WAIT TIME: 10 seconds after each question before taking answers. Cold-call 3 groups to share one similarity and one difference. Cold-call 3 learners to give one conversion and explain the steps. Ensure learners know	Structured Observation Table with Sort and Compare: recording the same attributes for each container lets learners see that height and opening are shared and only the shape differs. The unit revision supplies the tools needed to later describe capacity in litres.	Check that groups correctly record height 60 cm and opening radius 21 cm for all three containers and identify the bucket's smaller bottom. Observable indicators for units: learners convert $2,500 \text{ cm}^3$ to 2.5 L and 0.5 m^3 to 500 L correctly; learners can state the difference between volume and capacity in their own words. Note learners who confuse cm^3 with m^3 conversions (a common error: $1 \text{ m}^3 = 1,000 \text{ cm}^3$) for reteaching.

	They use a 1 L bottle or a cut-and-marked container as a reference for 1,000 cm ³ .	that h is the vertical height.		
Explain Phase  VIDEO: Visually Comparing the Capacity and Volume of Two Containers (Metric Units) Source: CK-12  Search ARES for similar videos  EXERCISE: Volume of cylinders, spheres, and cones word problems <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Groups return to their predictions. They compare their ranking with the tally and the observation table, then discuss: (1) Which features of the containers (height, opening, shape) might affect the amount of water held? (2) Why does 'same height and same opening' not guarantee 'same capacity'? Each group writes a short explanation, using the words volume and capacity correctly, on a sheet that will be kept. They also place a 'confidence' sticker (high, medium, low) beside their prediction. Learners use an everyday analogy: ugali served in a flat plate versus a pointed heap, or a maize heap in a sack versus a cone-shaped heap, to discuss how shape affects the amount of material.	Say: 'Turn to your group. Mama Chebet says the three containers are almost the same size. Is she right? Use evidence from your table.' Press for reasoning: 'What do you mean by size? Do you mean height, width, or space inside?' Distinguish size words: 'Is that volume or capacity?' WAIT TIME: 15 seconds after each question before calling on anyone. Cold-call 4 groups and ask the next group to restate the reasoning: 'Who can say that in your own words?' Do not tell learners which container is largest or reveal any ratio. Say: 'We don't have the mathematics to decide yet. What do we need to find out?' Record the learners' emerging needs on chart paper: 'How does a cone compare with a cylinder? What does the bucket's narrower bottom do?'	Claim-Evidence-Reasoning with Talk Moves (revoice, restate, add on): learners make a claim about the ranking, cite observations, and explain why. Talk moves ensure many voices contribute, and the unanswered question signals a gap in knowledge that drives the unit.	Listen for explanations that mention how shape narrows or tapers, not only height and width. Indicators: a learner states that 'same height and opening does not mean same volume'; uses volume and capacity correctly; identifies a missing mathematical tool. Note groups relying on 'looks bigger' with no reasoning and revisit them in Lesson 2.
Driving Question Board (DQB) Creation  VIDEO: Visually Comparing the Capacity and Volume of Three Containers (Metric Units) Source: CK-12  Search ARES for similar videos  EXERCISE: Apply volume of solids <i>Related topic: not an exact match</i>	The class posts the driving question on the DQB on the classroom wall: 'Why do the tea farmers' containers, a cone, a cylinder and a bucket, look nearly the same size but hold such different amounts of water, and how can we use mathematics to design a frustum-shaped container that holds exactly the capacity we need?' Each learner writes at least one wondering on a sticky note, for example: 'Does the cone really hold less, and how much less?', 'Why is the bucket in between?', 'How do I find the volume of a bucket shape?', 'Why is volume and capacity knowledge useful to a farmer?' Learners post notes	Say: 'Our big question is on the board. What are YOU wondering? Write one question on a sticky note, with your initials.' WAIT TIME: 10 seconds of silent thinking before writing. Cold-call 5 learners to read and post their notes aloud and ask: 'Which category does this belong in? Does anyone have a similar question?' Group similar questions with the class. Say: 'We will return to this board in every lesson, move questions to answered, and add new ones.' Highlight a learner question about usefulness and link to the KICD key inquiry: 'Why do farmers, traders and builders in Kenya need to know volume and capacity?' Do	Driving Question Board: learners' own wonderings are organised into categories, giving ownership and a visible record of growing understanding that will be revisited in each lesson.	Check that every learner has posted at least one relevant question. Indicators: questions that connect shape to volume (e.g., 'why is the cone smaller?') show conceptual curiosity, whereas only 'which is biggest?' questions suggest surface-level thinking. Record the question categories for later lessons. Note any learners who need support to phrase questions.

for this lesson. Source: Khan Academy (English - US curriculum) Search ARES for similar readings	under categories: Cone and cylinder, Bucket (frustum), Units and conversions, Designing our own container. Learners return to the board when they explain the key inquiry question: why is the knowledge of volume and capacity useful?	not answer questions yet.		
Model Building Phase  VIDEO: Comparing the Capacity and Volume of Three Containers (Metric Units) Source: CK-12 Search ARES for similar videos  HTML: Volume of Cylinders (Flexbook) <i>Related topic: not an exact match for this lesson.</i> Source: CK-12 Search ARES for similar readings	Learners help build the class model, a single diagram on chart paper that will grow each lesson. In groups they draw drum (A), cover (B) and bucket (C) side by side at the same scale, label the shared height (60 cm) and opening (42 cm diameter, 21 cm radius), and add a column 'Our prediction' with the class ranking and estimates. Beneath, they add a box 'Units we need' showing: volume in cm^3 and m^3; capacity in mL and L; $1 \text{ cm}^3 = 1 \text{ mL}$, $1000 \text{ cm}^3 = 1 \text{ L}$, $1 \text{ m}^3 = 1000 \text{ L}$. They add a box 'What we still need to find out' with questions from the DQB. Each learner copies the diagram in their notebook and completes a short exit ticket: 'Write your final prediction for the ranking and state one thing you are unsure about. Convert $3,500 \text{ cm}^3$ to litres and 2 m^3 to litres.'	Say: 'This model belongs to all of us. We will add to it every lesson. What must we draw so someone who missed class understands our puzzle?' Prompt: 'What is the same about the three containers? What is different? What do we not know yet?' Insist that actual capacities are left blank: 'We write only our predictions. We will find the real numbers by mathematics.' WAIT TIME: 10 seconds before collecting suggestions. Cold-call 3 learners to explain one part of the diagram. Collect exit tickets at the door. Preview: 'Next lesson we pour water to compare the cone and the cylinder. Think: how many times do you expect the cover to fit in the drum?'	Cumulative Class Model: a single shared diagram records what is known, predicted and unknown, and is revised in each lesson. This makes thinking visible and shows progress toward explaining the phenomenon.	Collect exit tickets. Indicators of success: (1) the diagram shows the same height and opening for all three containers; (2) conversions correct: $3,500 \text{ cm}^3 = 3.5 \text{ L}$ and $2 \text{ m}^3 = 2,000 \text{ L}$; (3) a final prediction with a reason and an honest statement of uncertainty. Learners who convert incorrectly receive a short unit-revision task before Lesson 2.

D. TEACHER REFLECTION

Which learners held the preconception that equal height and opening mean equal capacity, and how strongly? Did learners keep volume and capacity distinct, or did they use the words interchangeably? Which conversions caused errors (especially between m^3 and cm^3)? Did every learner post a genuine wondering, and did the DQB categories capture the main ideas? Were any predictions kept hidden because of fear of being wrong, and how can the next lesson make it safer to be wrong? Did I avoid revealing capacities or the one-third relationship? What should I adjust for Lesson 2, where learners pour water between the cone and the cylinder?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	We observed three containers from Mama Chebet's farm in Kericho: a cylindrical drum (A), a conical sisal cover (B) and a metal
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	frustum bucket (C), all about 60 cm tall with a 42 cm opening (21 cm radius), though their shapes are different. We recorded our predictions of which holds most.
What did I learn?	We learned that volume is the space inside a solid (cm^3 , m^3) and capacity is the amount a container can hold (mL, L), with $1 \text{ cm}^3 = 1 \text{ mL}$, $1000 \text{ cm}^3 = 1 \text{ L}$ and $1 \text{ m}^3 = 1000 \text{ L}$. We learned that equal height and opening do not tell us equal capacity.
How does this explain the phenomenon?	We cannot yet explain why the containers hold different amounts, but we predict that shape matters. We have posted our questions on the Driving Question Board and will use mathematics to test our predictions in the next lessons.

LESSON 2 (80 minutes (2 double lessons of 40 minutes each, or 1 double period)): Cone versus Cylinder: Why Does the Cover Hold Only One Third of the Drum?

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners discover by pouring that a cone fills a cylinder of the same base radius and height exactly three times, derive $V = \frac{1}{3} \pi r^2 h$, and use it to calculate the capacity of the Kericho conical sisal cover (B) in litres, comparing it with the cylindrical drum (A).
Knowledge	By the end of the lesson the learner should be able to determine the volume and capacity of a cone (Specific Learning Outcome a: determine the volume and capacity of cones and pyramids; this lesson covers cones only, pyramids come in Lesson 3). Learners state that $V(\text{cylinder}) = \pi r^2 h = 3 \times V(\text{cone})$ when r and h are equal, so $V(\text{cone}) = \frac{1}{3} \pi r^2 h$; they use $\pi = 22/7$ and the conversions $1 \text{ cm}^3 = 1 \text{ mL}$ and $1000 \text{ cm}^3 = 1 \text{ L}$. For the Kericho containers: drum A = $83,160 \text{ cm}^3 = 83.16 \text{ L}$ and cover B = 27.72 L .
Skills	Using a model cone and cylinder of the same base radius and height to establish the relationship between their volumes (KICD suggested learning experience); measuring r and h accurately; collecting and tabulating evidence; reasoning from repeated pours to a general formula; substituting into a formula; converting cm^3 to litres. NGSS-style practices embedded: Planning and carrying out investigations, Analyzing and interpreting data, Using mathematics and computational thinking, Constructing explanations, Developing and using models.
Attitudes	Critical thinking and problem solving: open-mindedness and creativity when discussing with peers and working out volume and capacity of containers. Unity: tolerance of different opinions while working in groups. Respect: listening to others' opinions while discussing the appropriate formula. Responsibility: taking care of containers and water. Environment and technology: using containers from the local environment to work out capacity.
Key Inquiry Question	Why is the knowledge of volume and capacity useful? (KICD key inquiry question). Lesson focus: Why does the conical cover (B) hold only one third of what the cylindrical drum (A) holds, even though both are 60 cm tall with a 42 cm opening?
Purpose in Storyline	Lesson 1 gave learners their predictions about drum A, cover B and bucket C, revised units of volume and capacity, and produced the DQB question about the drum versus the cover. Lesson 2 provides the first piece of mathematical evidence: it explains the cone-to-cylinder relationship and gives the exact capacities of A and B (83.16 L and 27.72 L ; ratio exactly 3). It does not yet explain the bucket (frustum, Lesson 4) or pyramids (Lesson 3). The one-third idea becomes the foundation for Lessons 3 to 5.
Safety Notes	Pour over trays or basins and mop spills immediately to prevent slipping. Use clean water only; do not drink it. Handle cut edges of card or plastic models with care. Use small clean containers or measuring cylinders to measure, and keep water away from electrical devices. Learners with mobility or visual difficulties are paired with a partner and given a role such as recorder or reader suited to their needs.

B. LESSON OVERVIEW

Mama Chebet's three containers are all 60 cm tall with a 42 cm opening, yet the drum (A) holds three times as much as the conical cover (B). In this lesson Ms. Wanjiru's class first revisits their Lesson 1 predictions and the DQB question about the drum versus the cover. Groups then investigate with a hollow cone and a hollow cylinder of the same base radius and height: they fill the cone to the brim and pour it into the cylinder, repeating until the cylinder is full, and record that exactly 3 cone-fills fill it. Learners analyse class data, explain why the result is general, and write $V(\text{cone}) = \frac{1}{3} \pi r^2 h$. They apply the formula to Kericho cover B ($r = 21 \text{ cm}$, $h = 60 \text{ cm}$, $\pi = 22/7$) to get $27,720 \text{ cm}^3 = 27.72 \text{ L}$, check it against drum A ($83,160 \text{ cm}^3 = 83.16 \text{ L}$), and verify the ratio of exactly 3. They update the DQB and add the cone-cylinder relationship to the cumulative class model, leaving bucket C and pyramids as open questions.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Volumes of cones intuition Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Volume of cones Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Learners recall Mama Chebet's containers and look at their Lesson 1 predictions on the wall. In pairs they look at a hollow model cone and a hollow model cylinder with the same base radius and the same height, which Ms. Wanjiru holds up. Each learner writes in their notebook: 'I predict the cylinder will hold ____ conefuls of water, because ____.' They also draw a quick sketch of the two solids and write their prediction as a number. This links back to the phenomenon: the drum (A) and cover (B) have the same height and opening, but different shapes.	Hold up the model cone and cylinder and say: 'These two are just like Mama Chebet's cover and drum. Same height, same opening. Before we pour a single drop, how many times do you think the cone must be filled and emptied to fill the cylinder?' Point to the DQB question from Lesson 1 and read it aloud. Give WAIT TIME of 10-15 seconds in silence while learners write. Then cold-call 4 learners (not volunteers) to share their number and reason, and record all predictions as tally marks on the board without saying which is right. Ask 2 further learners: 'What in the shape made you choose that number?' Say: 'Mr. Kiprop asks why his bucket is not as big as the drum. Today we start with the cone, and we will let the water tell us.' Ensure every learner has committed to a written prediction before moving on.	Predict-Commit-Share (making thinking visible): committing to a written numerical prediction before evidence creates a cognitive stake, so learners notice when the evidence agrees or disagrees with their ideas. Tally marks on the board turn private guesses into a class data set to revisit in the Explain phase.	Observe whether each learner has written a numerical prediction and a reason (check 100% of notebooks while circulating). Indicators: learners mention same height and same base but a narrowing shape; learners with 'twice' as an answer show the common belief that a cone is half a cylinder, to address in the Explain phase. Record the number of learners predicting 2, 3, 4 or other on the tally.
Observe Phase  VIDEO: Volume of a cone Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Volume of the Cone (PLIX) Source: CK-12  Search ARES for similar readings	In groups of 4 with roles (pourer, measurer, recorder, checker), learners first measure and record the base radius r and perpendicular height h of both the model cone and the model cylinder and confirm they are equal. They fill the cone to the brim with water over a tray, pour it into the cylinder, and count each pour until the cylinder is exactly full (3 conefuls), recording the water level after each pour (about $\frac{1}{3}$, $\frac{2}{3}$, full). They repeat the investigation twice to check and record results in a table: trial, conefuls needed, observations. Groups also fill the cylinder first and pour it into the	Say: 'Your job is to collect evidence, not to guess. Measure first: are the radius and height of your cone and cylinder really equal?' Circulate and ask: 'Where does the water level reach after the first pour? After the second?' Insist that the cone is filled exactly to the brim with the surface levelled: 'A full cone means level with the rim, not heaped.' Prompt groups that get 3 almost but not exactly with: 'What might cause a small difference? Spilled water? A cone not full to the brim?' Give WAIT TIME of 10-15 seconds after each question. Ask the recorder at 3 groups to report the number of	Structured investigation with repeated trials: pouring several times and recording in a table builds the idea that a pattern that holds across trials and groups is reliable evidence. Using a concrete model before the formula makes the abstract one-third idea physical and visible.	Check for observable indicators: (1) group measures r and h and confirms they are equal; (2) cone is filled to the brim and levelled before each pour; (3) group records the number of conefuls in the table for at least 2 trials; (4) group notices that the cylinder is full after exactly 3 pours. Note groups who report 'about 3' and ask them to explain sources of error. Use a quick tally of group results on the board.

	cone to see the overflow. Groups with larger demonstration containers may use a cone and cylinder with radius 7 cm and height 10 cm, so they can also read the water volume in mL (cone about 513 mL, cylinder about 1,540 mL using $\pi = 22/7$) to connect the pours to the Kericho figures. The thread to the phenomenon is explicit: A is the cylinder, B is the cone.	conefuls aloud. Remind learners to mop spills at once. Say: 'Mama Chebet's drum and cover are too big for the classroom, but they behave like these models. Keep that in mind.' Do not tell them the formula yet.		
Explain Phase  VIDEO: Volumes of cones intuition Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Volume of Cones (Flexbook) Source: CK-12  Search ARES for similar readings	Groups compare their results with the class tally and with their Predict-phase predictions, then discuss: 'What is the relationship between V(cylinder) and V(cone) when r and h are equal?' They write $V(\text{cylinder}) = 3 \times V(\text{cone})$, then rearrange to $V(\text{cone}) = 1/3 \times V(\text{cylinder}) = 1/3 \pi r^2 h$. Learners write the formula with symbols labelled (r = radius, h = perpendicular height, not the slant height). They then apply it to Kericho cover B: r = 21 cm, h = 60 cm, $\pi = 22/7$. Working: $V = 1/3 \times 22/7 \times 21 \times 21 \times 60 = 1/3 \times 22/7 \times 26,460 = 1/3 \times 83,160 = 27,720 \text{ cm}^3$. Converting: $27,720 \div 1000 = 27.72 \text{ L}$. They also calculate drum A: $22/7 \times 21 \times 21 \times 60 = 83,160 \text{ cm}^3 = 83.16 \text{ L}$, and check the ratio $83.16 \div 27.72 = 3$. Learners write a short explanation to Mr. Kiprop and Mama Chebet: 'The cover holds one third of the drum because...'. Extension: learners predict what would happen to the capacity of the cover if the height were doubled but the radius stayed the same (it doubles) and why that does not make a cone hold the same as the drum.	Ask: 'Looking at all our data, what stays the same in every group?' Give WAIT TIME of 10-15 seconds, then cold-call 3 learners. Say: 'Some of you predicted 2. What did the evidence say? What was your thinking, and what would you say now?' Welcome revised ideas: 'Changing your mind because of evidence is exactly what scientists and mathematicians do.' Write $V(\text{cylinder}) = \pi r^2 h$ on the board and ask: 'If the cone fits into it 3 times, what is the cone's volume?' Cold-call 2 learners to complete the formula. Model the first substitution into the formula using the cover B data with a think-aloud, showing that $21 \times 21 = 441$ and $22/7 \times 441 = 1,386$ before multiplying by 60 and dividing by 3. Then let pairs do the drum A calculation independently. Stress units: 'cm³ is volume, litres is capacity. $1000 \text{ cm}^3 = 1 \text{ L}$.' Ask 3 learners to explain the conversion in their own words. Check that everyone uses perpendicular height. Ask: 'What does the ratio of exactly 3 tell Mama Chebet about her drum and cover?' Do not discuss the bucket: say 'The bucket is a different shape; we will come back to it in Lesson 4.'	Claim-Evidence-Reasoning (CER): learners state a claim (cone volume is one third of cylinder volume), support it with evidence from the pouring data, and give reasoning linking it to the formula and the Kericho containers. Number talk and a worked example with think-aloud model how to move from an experimental relationship to a formula and then to a real capacity.	Collect each learner's written CER statement and calculation on an exit slip. Indicators: (1) states the 3-to-1 relationship using evidence; (2) writes $V = 1/3 \pi r^2 h$ correctly; (3) substitutes r = 21 and h = 60 correctly with $\pi = 22/7$; (4) gets $27,720 \text{ cm}^3$ and converts to 27.72 L; (5) verifies the ratio $83.16 \div 27.72 = 3$. Common errors to note: using diameter 42 instead of radius 21, forgetting to divide by 3, using slant height, or dividing by 100 instead of 1000 in the conversion. Provide immediate individual feedback.

Driving Question Board (DQB) Creation  VIDEO: Volume of a cone Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Volume of Cones Practice Source: CK-12  Search ARES for similar readings	Learners return to the class DQB posted on the wall. In groups they write on sticky notes (a) what the pouring investigation answered, for example 'Why does the cover hold less than the drum?' with the answer 'one third because $V(\text{cone}) = \frac{1}{3} \pi r^2 h$', and (b) new questions, for example 'Does the one-third idea work for a pyramid?' or 'The bucket is wider at the top than the bottom, so how do we find its volume?' or 'What is the capacity of a cone-shaped water tank?' They place the answered note next to the original question and add new questions in a 'still wondering' area, linking them to Mr. Kiprop's question about the bucket.	Say: 'Look at the DQB question we posted in Lesson 1 about the drum and the cover. Can we move it to the answered section? What evidence do we have?' Give WAIT TIME of 10-15 seconds and cold-call 2 groups to justify. Then say: 'Now what new questions has the cone raised for you? Write at least one on a sticky note.' Circulate and push for questions that connect to the unit: 'How could this help us with Mr. Kiprop's bucket?' Ask 3 learners to read their questions aloud and group similar ones together. Point out that the bucket and pyramid questions will be dealt with in Lessons 3 and 4 without giving the answers.	Driving Question Board as a tracking tool: connecting new evidence to the original questions and generating new ones shows learners that understanding grows in steps and gives them ownership of the unit's direction.	Check each group's sticky notes: Does the answered note cite the pouring evidence and the formula? Do new questions link to frustums, pyramids, or real contexts? Indicators of progress: questions move from 'which is bigger?' to 'how can we calculate or design?'. Record the number of groups posting at least one new mathematically focused question.
Model Building Phase  VIDEO: Metric system: units of volume <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Volume of Cones: Strawberry Ice Cream (PLIX) Source: CK-12  Search ARES for similar readings	Learners add to the single cumulative class diagram that began in Lesson 1. Each group draws or adds labelled sketches of drum A (cylinder, $r = 21$ cm, $h = 60$ cm, $V = \pi r^2 h = 83.16$ L) and cover B (cone, $r = 21$ cm, $h = 60$ cm, $V = \frac{1}{3} \pi r^2 h = 27.72$ L), with an arrow and the label '$\times \frac{1}{3}$' or 'cylinder = 3 cones'. They leave a space for bucket C marked with a question mark, and a note, 'between cone and cylinder? Lesson 4 will test'. Learners compare the model with their Lesson 1 predictions and note one thing they got right and one thing to revise. Finally, in their notebooks each learner writes a two-sentence answer to the question, 'Why is knowing how to calculate volume and capacity useful for Mama Chebet?'.	Say: 'Our model is one diagram that grows each lesson. What should we add today?' Give WAIT TIME of 10-15 seconds, then cold-call 3 learners to add one labelled element each. Ask: 'What does the model now explain about the phenomenon, and what does it not yet explain?' Cold-call 2 learners. Make sure the diagram shows A and B with equal r and h, and the exact litres: 83.16 L and 27.72 L. Say: 'Notice the bucket is still a question mark. A model is never finished until it explains everything.' Close by asking one learner to summarise the day's learning in one sentence, and one learner to say how it helps Mama Chebet and Mr. Kiprop. Preview Lesson 3: 'Next time we test whether the one-third idea works for a pyramid.'	Cumulative model building and revision: adding to a single class diagram each lesson shows learners how evidence refines a model, and marking what is still unexplained keeps the driving question alive.	Check the class model and learners' notebooks for: correct labels ($r = 21$ cm, $h = 60$ cm for both), the relationship 'cylinder = 3 $\times$ cone', capacities 83.16 L and 27.72 L, and an honest placeholder for the bucket. Check the two-sentence reflection for a valid use of volume and capacity (for example choosing containers, estimating how much tea leaves or water a container can hold). Note learners who still confuse volume and capacity or radius and diameter for re-teaching.

D. TEACHER REFLECTION

After the lesson, reflect: (1) Did every group obtain 3 cone-fuls, and how did I help groups whose result was not exactly 3 to see the cause (spilling, partial filling) without giving the answer? (2) How many learners changed their prediction from 2 to 3, and did I value that change? (3) Which common errors appeared (diameter instead of radius, forgetting $\frac{1}{3}$, cm^3 to L conversion), and which learners need targeted support before Lesson 3? (4) Was WAIT TIME of 10-15 seconds used consistently, and did cold-calling include quieter learners? (5) Did the DQB reflect the learners' own questions about the bucket and pyramids? (6) Were safety and resource-use routines (spill control, caring for models) followed? Plan adjustments for Lesson 3, which builds on the one-third idea.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	We filled a model cone to the brim and poured it into a cylinder of the same base radius and height. The cylinder was full after exactly 3 cone-fuls, in every trial and every group. For the Kericho containers, drum A holds 83.16 L and cover B holds 27.72 L, a ratio of exactly 3.
What did I learn?	The volume of a cone is one third of the volume of a cylinder with the same base radius and height: $V = \frac{1}{3} \pi r^2 h$. With $\pi = \frac{22}{7}$, $r = 21$ cm, $h = 60$ cm, $V = 27,720$ cm^3 , and 1000 $\text{cm}^3 = 1$ L gives a capacity of 27.72 L.
How does this explain the phenomenon?	The conical cover (B) and cylindrical drum (A) look alike because both are 60 cm tall with a 42 cm opening, but the cone narrows to a point, so it holds exactly one third of the drum: 27.72 L against 83.16 L. We still need to find out why the bucket (C) holds an amount in between.

LESSON 3 (80 minutes (2 x 40-minute lessons)): Volume of Pyramids: Does the One-Third Pattern Hold Beyond Cones?

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners establish by pouring evidence that $V(\text{pyramid}) = \frac{1}{3} \times \text{base area} \times \text{height}$, apply it to the pyramid grain-store model, and convert volume to capacity. They connect the result to the Lesson 2 one-third pattern for cone and cylinder and test their Lesson 1 predictions.
Knowledge	a) Determine the volume and capacity of cones and pyramids (pyramid focus in this lesson). Volume of a pyramid $V = \frac{1}{3} \times \text{base area} \times \text{height}$, from the 3-to-1 relation between a cuboid and a pyramid of equal base and height. Grain-store model: square base side 30 cm, $h = 40$ cm, cuboid $36,000 \text{ cm}^3$, pyramid $12,000 \text{ cm}^3 = 12 \text{ L}$. Conversions: $1 \text{ cm}^3 = 1 \text{ mL}$; $1000 \text{ cm}^3 = 1 \text{ L}$. h is always the perpendicular height, never the slant height. Use $\pi = \frac{22}{7}$ where needed.
Skills	Measuring base side and perpendicular height; pouring and comparing volumes; calculating base area; applying a formula; converting cm^3 to L; communicating reasoning (NGSS SEP: Planning and Carrying Out Investigations; Analyzing and Interpreting Data; Constructing Explanations; Developing and Using Models; Using Mathematics and Computational Thinking).
Attitudes	Core competencies: Critical thinking and problem solving (open-mindedness and creativity when working out volume and capacity of containers); Creativity and imagination (connection skills). Values: Respect (listening to others' opinions while discussing formulas); Unity (tolerance of different opinions in groups). PCI: Environment and technology (using appropriate containers and locally available materials from the environment). Appreciate volume and capacity in real-life situations such as grain storage in the Rift Valley.
Key Inquiry Question	Why is the knowledge of volume and capacity useful?
Purpose in Storyline	Lesson 2 showed that the cone holds one third of the cylinder (cover B 27.72 L against drum A 83.16 L). This lesson asks whether the one-third pattern is special to circles or also holds for pyramids, so learners can later explain the bucket (Lesson 4) and design frustums (Lessons 4-5). It also revisits Lesson 1 predictions and updates the DQB and class model.
Safety Notes	Use dry sand, rice or maize grain for the pour test if water is scarce, and keep the floor dry to prevent slipping. Handle card or plastic hollow models carefully to avoid sharp edges; trim edges before use. Wipe up spills immediately. Wash hands after handling grain. Do not eat the grain used in experiments.

B. LESSON OVERVIEW

Ms. Wanjiru opens by recalling Mama Chebet's containers and the Lesson 2 finding that the conical cover B holds exactly one third of drum A. She asks: 'Is the one-third pattern only about circles?' Learners predict how many pyramid-fulls of grain fill a cuboid with the same square base and height, then pour to test it with hollow models (base 30 cm x 30 cm, $h = 40$ cm). They find 3 pyramids fill the cuboid exactly, build the rule $V = \frac{1}{3} \times \text{base area} \times \text{height}$, and apply it to the pyramid grain-store model ($12,000 \text{ cm}^3 = 12 \text{ L}$, cuboid $36,000 \text{ cm}^3$). They compare with their Lesson 1 predictions, discuss why the pyramid is not just a different-looking cone, update the DQB and extend the class model with a pyramid and cuboid pair.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase	Learners recall the Lesson 2 result	Say: 'Last lesson, Mama Chebet's	Prediction with reasoning, plus	Observable indicators: at least 8 of

 VIDEO: Volume of pyramids intuition Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Volume of prisms and pyramids Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	(cone holds one third of cylinder with the same base and height: cover B 27.72 L, drum A 83.16 L). Ms. Wanjiru shows a hollow cuboid and a hollow square-based pyramid of equal base (30 cm x 30 cm) and height (40 cm), described as models of a maize store in the Rift Valley. In pairs they write: 'How many pyramids of grain will fill the cuboid? Why do you think so?' and sketch their reasoning. They also glance back at their Lesson 1 predictions about the three Kericho containers and mark whether the same shape logic seems to apply.	cover held exactly one third of her drum. Look at these two models. Same base, same height. Write down how many pyramids of maize will fill this cuboid, and give a reason.' Give WAIT TIME of 12 seconds of silent thinking before pairs talk. Cold-call 4 pairs to share without evaluating: 'What number did you choose and what made you choose it?' Tally predictions on the board (e.g., 2, 3, 4, 'more than 4'). Ask: 'Who changed their idea after hearing a classmate? What convinced you?' Do not reveal the answer. Circulate and note learners who link their answer to Lesson 2.	transfer from a prior pattern: asking learners to commit to a number and justify it surfaces whether they transfer the one-third idea from cone/cylinder to pyramid/cuboid or guess. The tally makes disagreement visible and creates a need to test with evidence.	30 learners cite the Lesson 2 one-third idea in their reasoning; each pair records a number plus a reason; teacher notes misconceptions such as 'a pyramid is half the cuboid because it has a point'. Record the tally for comparison after the pour test.
Observe Phase  VIDEO: Volume of a Pyramid - Overview Source: CK-12  Search ARES for similar videos  EXERCISE: Volume of prisms and pyramids Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	In groups of 4, learners receive a hollow cuboid and a hollow square-based pyramid with the same base and height (base 30 cm x 30 cm, perpendicular height 40 cm), plus dry sand or maize grain, a tray, a metre rule and a measuring jug. Roles: measurer, pourer, recorder, checker. Step 1: measure the base side and perpendicular height of both models and record them. Step 2: fill the pyramid level to the brim and pour it into the cuboid; count how many pyramid-fulls fill the cuboid exactly. Step 3: repeat once to confirm. Step 4: calculate the cuboid volume $30 \times 30 \times 40 = 36,000 \text{ cm}^3$ and compute $36,000 / 3 = 12,000 \text{ cm}^3$ for the pyramid. Step 5: fill the pyramid with water or grain and pour into the jug to check it is about 12 L in the full-size version or scaled to model capacity; record the ratio. Step 6: write a table of cuboid volume, pyramid volume and ratio.	Say: 'Level every pour with a ruler. A heaped pyramid will give you a false answer. Measure the vertical height from the apex straight down to the base, not along the sloping face.' Demonstrate measuring perpendicular height once with a set square. While circulating, ask each group: 'What does your third pour show you?' and 'Is your pyramid level? How do you know?' Allow WAIT TIME of 10 seconds after each question. Cold-call 3 groups to report their counts and compare with the prediction tally. Prompt: 'If one group got 2.8 pours, what might have caused the difference?' (spillage, heaping, uneven fill). Remind groups of the safety notes and spill clean-up.	Data collection and pattern recognition: learners collect their own measurements and counts, then compare group results to see a stable 3-to-1 ratio despite small errors. Linking the table to the Lesson 2 cone/cylinder ratio lets learners see the same ratio in a new solid pair.	Observable indicators: groups level pours accurately; measurements of base side and perpendicular height are recorded in cm; the table shows cuboid $36,000 \text{ cm}^3$, pyramid $12,000 \text{ cm}^3$ and ratio 3 : 1; learners can say why slant height was not used. Checklist: 3 of 3 pours counted, units recorded, ratio stated.
Explain Phase	Learners use their evidence to	Say: 'Use only your pouring	Constructing an explanation from	Observable indicators: learners

 VIDEO: Volume of pyramids intuition Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Apply volume of solids <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	construct the rule. First, each group writes 'cuboid volume = base area x height' and 'pyramid = 1/3 of the cuboid', then combines them into $V = \frac{1}{3} \times \text{base area} \times \text{height}$. Next they apply it to the grain-store model: base area = $30 \times 30 = 900 \text{ cm}^2$, $V = \frac{1}{3} \times 900 \times 40 = 12,000 \text{ cm}^3$, capacity = $12,000 / 1000 = 12 \text{ L}$. They then answer: (a) how many litres would the matching cuboid store hold? (36 L) (b) if a maize store pyramid has base side 60 cm and h 80 cm, how does the volume change? Groups compute $\frac{1}{3} \times 3,600 \times 80 = 96,000 \text{ cm}^3 = 96 \text{ L}$. They write a short explanation of why the pyramid holds one third of the cuboid, then a one-sentence comparison with the cone-cylinder result from Lesson 2. They also revisit Lesson 1 predictions: Which predictions about shape and volume were supported by the pyramid evidence?	evidence. What does the data tell you about the pyramid and the cuboid?' Allow WAIT TIME of 15 seconds, then cold-call 4 learners to state the rule in their own words and 2 more to critique wording: 'What would you add so the rule works for any pyramid?' Insist on 'perpendicular height'. Ask: 'In both cone/cylinder and pyramid/cuboid, what is the same about the two solids?' (same base, same height) and 'What stays the same in the ratio?' (3 to 1). Do not claim a proof; explain that the pouring is evidence consistent with the formula. Ask: 'Why is this formula useful to a farmer who wants to store maize?' Check conversions: 'How many mL are in $12,000 \text{ cm}^3$?' Address misconception: 'A taller pyramid with a bigger base still follows the same rule.'	evidence (claim-evidence-reasoning): learners state the claim $V = \frac{1}{3} \times \text{base area} \times \text{height}$, cite the 3-to-1 pour data as evidence, and reason about equal base and height. Comparing with Lesson 2 builds a generalisation: pyramids and cones are one third of the prism or cylinder with the same base and height.	write the formula with base area and perpendicular height; compute 900 cm^2 then $12,000 \text{ cm}^3$ then 12 L without error; convert cm^3 to L correctly; give a CER explanation linking 36,000 to 12,000. Exit check: calculate the volume and capacity of a pyramid with base side 30 cm and height 40 cm (answer $12,000 \text{ cm}^3 = 12 \text{ L}$) and justify each step. Note learners using slant height for follow-up.
Driving Question Board (DQB) Creation  VIDEO: Volume of a Pyramid - Overview Source: CK-12  Search ARES for similar videos  EXERCISE: Volume by multiplying area of base times height <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English	Learners return to the class DQB posted on the wall. In groups they (1) place a sticky note answering the Lesson 2 question about the drum versus the cover with 'Answered: 3 to 1, because cone = 1/3 cylinder' if not yet done, (2) mark any questions now answered by the pyramid evidence, e.g. 'Does the one-third rule work for other solids?', and (3) write at least one new question, e.g. 'What happens when the top of a cone is cut off, like Mr. Kiprop's bucket?' or 'How do we find the volume of a frustum of a pyramid?' Each group posts one new question and sorts it under Cone, Pyramid, Frustum or Design.	Say: 'Go back to Mama Chebet's three containers. What have we answered, and what do we still not know about the bucket?' Give WAIT TIME of 10 seconds. Cold-call 3 groups to read a new question aloud and say which evidence prompted it. Ask: 'Mr. Kiprop asks why his bucket is not as big as the drum. Which question on our board helps us with that?' Point out that the pyramid is a stepping stone toward the frustum. Cluster questions and mark 'Lesson 4: frustum of a cone' next to the bucket question. Do not answer frustum questions yet.	Question-driven learning with a public record: the DQB tracks what the class now knows and what it still needs, motivating the next lessons and giving learners ownership of the inquiry path toward the bucket.	Observable indicators: each group posts at least one new, relevant question; at least one question links to the frustum or the bucket; questions are sorted correctly; learners can name the evidence that answered an old question. Teacher records which groups ask only recall questions versus extension questions.

- US curriculum) Search ARES for similar readings				
Model Building Phase  VIDEO: Volumes of cones intuition <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum) Search ARES for similar videos  EXERCISE: Volume as area of base times height <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum) Search ARES for similar readings	Learners add to the cumulative class diagram started in Lessons 1-2. On the wall poster, each group draws a pyramid and cuboid pair (base 30 cm x 30 cm, h 40 cm) beside the cone and cylinder pair, labels the volumes (36,000 cm³ and 12,000 cm³ = 12 L) and writes the generalised rule 'solid with a point = $\frac{1}{3}$ x base area x height'. They add an arrow from Mama Chebet's cover B (27.72 L) to the pyramid grain store to show the shared one-third idea and write a question mark by the bucket C (frustum). Individually, learners then write in their notebooks: 'The one-third rule works for ___ and ___ because _____. I still wonder _____. They also compare their Lesson 1 prediction about the cover B and note whether their reasoning would have been better with the one-third idea.	Say: 'Our model must show what we now know and what is still unknown. What belongs on the poster?' Give WAIT TIME of 12 seconds. Cold-call 3 learners to explain one label and why it is correct, and ask 2 others: 'What would you change?' Prompt: 'Where on this diagram would the bucket go? What do we need to find out?' Check that labels use perpendicular height and units. Ask learners to share one place where they could use the pyramid rule in Kenya (maize stores in the Rift Valley, roof designs, heaps of tea leaves or sand).	Developing and revising a model: the single cumulative diagram integrates cone/cylinder with pyramid/cuboid so learners see a unifying idea. Marking the bucket as the next unknown keeps the phenomenon central and shows that models are revised as evidence grows.	Observable indicators: the diagram shows correct values (36,000 cm³, 12,000 cm³, 12 L) and the $\frac{1}{3}$ rule; units and the term perpendicular height are used; the notebook sentence links the rule to both cone and pyramid; the bucket is marked as unresolved. Collect notebooks to identify learners needing support on base area or unit conversion.

D. TEACHER REFLECTION

Did learners level their pours and measure perpendicular height rather than slant height? How many cited the Lesson 2 one-third idea in their predictions, and did their ideas change after the pour? Which learners confused volume (cm³) with capacity (L)? Did the new DQB questions point toward the frustum bucket? Plan to revisit any learners using slant height or calculating base area incorrectly before Lesson 4, and consider more sand or grain if pours were messy.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	We observed that a hollow square-based pyramid (base 30 cm x 30 cm, height 40 cm) filled a cuboid of the same base and height exactly 3 times, so the cuboid held 36,000 cm ³ and the pyramid 12,000 cm ³ .
What did I learn?	We learned that the volume of a pyramid is $V = \frac{1}{3} \times \text{base area} \times \text{perpendicular height}$, and that 12,000 cm ³ is 12 L. The same one-third relationship we found for the cone and cylinder holds for pyramids and cuboids.
How does this explain the	We can explain that a pyramid holds one third of the cuboid with the same base and height, just as Mama Chebet's cover B holds

phenomenon?	one third of drum A. This moves us toward explaining the bucket (a frustum), which we will investigate next.
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LESSON 4 (80 minutes (2 x 40-minute lessons)): Frustum of a Cone: Cutting the Bucket from a Cone

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners derive the formula for the volume of a frustum of a cone by subtracting a small cone from a large cone, use similar triangles to find the missing height, and calculate the capacity of Mr. Kiprop's bucket (C) so that they can explain why the bucket holds less than the drum (A) and more than the cover (B).
Knowledge	By the end of the lesson, the learner should be able to: (a) compute volume and capacity of a frustum of a cone (SLO b). Learners know that a frustum is the solid left when the top of a cone is cut off parallel to the base; that $V(\text{frustum}) = V(\text{large cone}) - V(\text{small cone})$; that similar triangles give the missing height of the small cone; and that $V = \frac{1}{3} \pi h (R^2 + Rr + r^2)$. With $\pi = \frac{22}{7}$, bucket C ($R = 21$ cm, $r = 14$ cm, $h = 60$ cm) holds $58,520 \text{ cm}^3 = 58.52 \text{ L}$, between cover B (27.72 L) and drum A (83.16 L).
Skills	Measuring radii and perpendicular height with rulers and tape; using similar triangles and ratios to find the missing height (SEP: Using mathematics and computational thinking); algebraic derivation by factorising; unit conversion cm^3 to L; constructing explanations from evidence (SEP: Constructing explanations); developing and revising a model (SEP: Developing and using models).
Attitudes	Core competencies: Critical thinking and problem solving (open-mindedness and creativity when discussing with peers and working out the volume and capacity of containers); Creativity and imagination (connection skills when working out the volumes of frustums from cones). Values: Unity (tolerance of different opinions while working with peers); Respect (listening to others' opinions while discussing appropriate formulas). PCI: Environment and technology (using appropriate containers from the local environment to work out capacity).
Key Inquiry Question	Why is the knowledge of volume and capacity useful?
Purpose in Storyline	Lesson 4 is the turning point of the storyline. Lesson 2 gave the cone formula (cone = $\frac{1}{3}$ of cylinder; drum A 83.16 L , cover B 27.72 L) and Lesson 3 extended the one-third idea to pyramids. Learners now use these ideas to resolve the anchoring phenomenon: bucket C is a large cone with a small cone removed, so it holds 58.52 L , between cover B and drum A. It also sets up Lesson 5 (frustum of a pyramid), which reuses the large-minus-small method.
Safety Notes	Use a plastic or smooth-edged bucket; check the rim for sharp metal edges. Wipe up spilled water immediately to prevent slipping. Use scissors and card for the paper cone cut-outs under supervision. Learners carry containers carefully and share the measuring tools fairly.

B. LESSON OVERVIEW

Ms. Wanjiru brings in a bucket like the one Mr. Kiprop carries to the Kericho weighing station and asks, 'Mr. Kiprop asked Mama Chebet why his bucket is not as big as her drum. Can we answer him with mathematics?' Learners first revisit their Lesson 1 predictions about bucket C. They then measure a real frustum bucket (top radius R , bottom radius r , height h) and, using a paper cone model, imagine extending the sides until they meet at a point to form the full cone. Using similar triangles, they find the missing height (small cone $h = 120$ cm, full cone $h = 180$ cm for bucket C). They subtract the small cone ($24,640 \text{ cm}^3$) from the large cone ($83,160 \text{ cm}^3$) to get $58,520 \text{ cm}^3 = 58.52 \text{ L}$, then generalise to $V = \frac{1}{3} \pi h (R^2 + Rr + r^2)$. Finally, they compare the bucket with drum A and cover B, resolve the phenomenon, add the bucket to the class model, and update the DQB.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Volume of a cone Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Solve similar triangles (basic) Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Learners look at the class bucket beside pictures of drum A and cover B (all about 60 cm tall, 42 cm opening). They look back at their Lesson 1 predictions and individually write: (1) Will bucket C hold more or less than cover B? More or less than drum A? (2) Sketch the bucket and extend its slanting sides upwards and downwards until they meet. What shape do you see? Then they use Lessons 2 and 3 to predict how the cone and the one-third idea might help find the bucket's volume. They share in pairs and then record a class prediction: 'The bucket is between cover B and drum A, but how much?'	Greet: 'Mr. Kiprop carries leaves in this bucket and asks why it is not as big as Mama Chebet's drum. Let us find out.' Hold up the bucket. Ask: 'Look at your Lesson 1 prediction. Where did you place the bucket?' Say: 'Take 2 minutes to write your answer silently. Do not talk yet.' Give WAIT TIME of 10-15 seconds after each question. Ask: 'If I carried on the slanting sides of this bucket until they meet, what solid would I get?' Cold-call 4 learners to share sketches. Ask: 'In Lesson 2 we learnt that cone volume is one third of cylinder volume. How could that help us with a bucket that is not a cone?' Cold-call 3 more. Record ideas without judging; do not give the answer.	Prediction with revisiting prior ideas (Think-Pair-Share): returning to Lesson 1 predictions makes learners' thinking visible, exposes the misconception that the bucket must be about half of the drum or equal to the cone, and creates a need for a method. Sketching the extended sides starts the 'missing cone' idea that the Explain phase builds on.	Observe whether learners (a) place the bucket between cover B and drum A, (b) sketch the extended sides meeting at a point to form a cone, and (c) link Lesson 2's cone formula to the problem. Note learners who predict by guessing or who say the bucket is 'half of the drum'; ask them for reasons. Tally how many correctly order the three containers. Look for correct vocabulary (frustum, cone, radius) in written work.
Observe Phase  VIDEO: Volume of a cone Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Solve similar triangles (advanced) Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	In groups of 4, learners take measurements. Task 1: measure the top diameter (42 cm) and bottom diameter (28 cm) of the class bucket, or of the data card for Mr. Kiprop's bucket C, and the perpendicular height (60 cm); calculate $R = 21$ cm and $r = 14$ cm. Task 2: using a paper cone model that has the top cut off parallel to the base, they extend the slanting sides to find the missing small cone and draw a cross-section of the side view with $R = 21$, $r = 14$, $h = 60$ and the unknown height x of the small cone. Task 3: they notice the two triangles have the same shape (similar) and record the ratio $r : R = 14 : 21 = 2 : 3$. They set up $x / (x + 60) = 14/21 = 2/3$, cross-multiply ($3x = 2x + 120$) and solve $x = 120$ cm. They record the full cone height = $120 + 60 = 180$ cm. Task 4: they calculate the large	Distribute buckets, tape measures, rulers, paper cone models, calculators and data cards. Say: 'Measure the diameter across the widest part, then halve it for the radius. Measure height straight up, not along the slanting side.' Circulate and ask: 'Where is your tape measure? Is it along the vertical?' Pose the key question: 'If we continue the slanting sides, where do they meet? How tall is the small cone we cut off?' Give WAIT TIME of 10-15 seconds before offering help. Prompt: 'Which two triangles have the same shape? How do you know?' Ask groups to name the ratio of their sides. Cold-call 3 groups to share the value of x and how they found it. Stop the class: 'Everyone check: is the big cone 180 cm tall? Show with your fingers: yes (thumb up) or no.' Watch for	Gathering evidence with structured data recording and scaffolded problem-solving (Using Mathematics and Computational Thinking): measuring the real bucket, building the cross-section diagram and applying similar triangles let learners see the frustum as a big cone minus a small cone, so the idea comes from evidence and not from being told. Doing the calculation step by step builds procedural fluency and uses the cone formula from Lesson 2.	Observable indicators: (a) learners measure $R = 21$ cm, $r = 14$ cm and $h = 60$ cm within about 1 cm; (b) the cross-section diagram labels R , r , h and the unknown x correctly; (c) the equation $x / (x + 60) = 2/3$ is set up and solved to $x = 120$; (d) volumes of large cone $83,160 \text{ cm}^3$ and small cone $24,640 \text{ cm}^3$ are correct; (e) the subtraction and conversion to 58.52 L are correct. Use a quick check: ask each group to display the full cone height and the bucket capacity on mini-whiteboards. Note groups that use the slant height or forget to halve the diameter.

	cone ($r = 21$, $h = 180$) $V = \frac{1}{3} \times 22/7 \times 441 \times 180 = 83,160 \text{ cm}^3$ and the small cone ($r = 14$, $h = 120$) $V = \frac{1}{3} \times 22/7 \times 196 \times 120 = 24,640 \text{ cm}^3$, then subtract: $83,160 - 24,640 = 58,520 \text{ cm}^3$. They convert: $58,520 \text{ cm}^3 = 58.52 \text{ L}$. Optional check: pour 58.52 L equivalent in 1 L containers or measure the data card.	learners who use the slant height instead of the perpendicular height; correct this with the paper cone model.		
Explain Phase  VIDEO: Volumes of cones intuition <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Volume Of A Cone Source: math_expression  Search ARES for similar readings	Learners explain the pattern generally. Step 1: they write $V(\text{frustum}) = V(\text{large cone}) - V(\text{small cone})$ in letters, with the small cone height x and the full cone height $H = x + h$: $V = \frac{1}{3} \pi R^2 (x + h) - \frac{1}{3} \pi r^2 x$. Step 2: with guidance, they use similar triangles $x / (x + h) = r / R$ to show $x = rh / (R - r)$ (for bucket C: $14 \times 60 / 7 = 120 \text{ cm}$, which matches). Step 3: they substitute and factorise, using $R^3 - r^3 = (R - r)(R^2 + Rr + r^2)$, to get $V = \frac{1}{3} \pi h (R^2 + Rr + r^2)$. They then test the formula on bucket C: $\frac{1}{3} \times 22/7 \times 60 \times (441 + 294 + 196) = \frac{1}{3} \times 22/7 \times 60 \times 931 = 58,520 \text{ cm}^3 = 58.52 \text{ L}$, the same as the subtraction method. Step 4: they compare the three containers in a table: cover B 27.72 L, bucket C 58.52 L, drum A 83.16 L. They explain in writing: all three are about 60 cm tall with a 42 cm opening, but shape matters. The cover tapers all the way to a point, the bucket tapers only to a 28 cm base, and the drum does not taper. They compute that the bucket is about 70% of the drum ($58.52 / 83.16 = 0.70$) and about 2.11 times the cover. They write an answer to Mr. Kiprop's question and give a CER (claim, evidence, reasoning) response.	Ask: 'What does the word frustum tell you about how the bucket is made?' Write $V = \text{large cone} - \text{small cone}$ on the board and ask learners to write it in letters. Say: 'Turn to your partner and explain why subtracting works. You have 1 minute.' Give WAIT TIME of 10-15 seconds, then cold-call 4 pairs. For the algebra, scaffold: 'What do R^3 and r^3 have in common? Remember the factorisation of a difference of cubes.' If learners struggle, show the pattern with numbers (441, 294, 196) before the letters. After the formula, ask: 'Which method do you prefer for the bucket, subtracting or the formula? When would each be useful?' Cold-call 3 learners. Then ask: 'Mama Chebet's drum holds 83.16 L, her cover 27.72 L and Mr. Kiprop's bucket 58.52 L. Explain in a sentence for Mr. Kiprop why his bucket holds less than the drum.' Select 3 volunteers and 2 cold-calls to share CER answers. Highlight the key point: same height and opening do not mean same volume; shape matters. Correct the misconception that 'the bucket is half the drum' with the 70% value.	Claim-Evidence-Reasoning (CER) with a general derivation: moving from the specific numbers of bucket C to the letters R, r and h shows learners that mathematics lets us predict capacities for any frustum. Comparing the three containers in a table makes the final resolution of the phenomenon concrete. Peer explanation (Think-Pair-Share) helps learners put the reasoning into their own words and listen to different views (Respect and Unity values).	Observable indicators: (a) learners write $V(\text{frustum}) = V(\text{large cone}) - V(\text{small cone})$ correctly; (b) they state that $x = rh / (R - r)$ or can find the missing height with ratios; (c) they use the formula $\frac{1}{3} \pi h (R^2 + Rr + r^2)$ correctly with $R = 21$, $r = 14$, $h = 60$ and obtain 58.52 L; (d) the CER response contains a claim (bucket holds 58.52 L, between B and A), evidence (the numbers) and reasoning (shape: large cone minus small cone). Check mini-whiteboards for a one-line explanation of why the bucket holds less than the drum. Exit question: 'A bucket has $R = 15 \text{ cm}$, $r = 10 \text{ cm}$ and $h = 24 \text{ cm}$. Write the expression you would calculate.' ($\frac{1}{3} \times 22/7 \times 24 \times (225 + 150 + 100)$).
Driving	Learners return to the class DQB	Say: 'Walk to the DQB with your	Driving Question Board as a living	Observable indicators: (a) learners

Question Board (DQB) Creation  VIDEO: Metric system: units of volume <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Solve similar triangles (basic) Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	posted on the classroom wall. In groups, they (1) find and mark the questions that today's work has answered, for example 'Why does the bucket hold less than the drum but more than the cover?' and 'How do you find the volume of a bucket-shaped container?'; (2) write the answers on sticky notes with evidence such as 58.52 L and the formula; (3) write at least one new question that has come from today's work, for example 'Does the same large-minus-small method work for a frustum of a pyramid, like a maize bin?' or 'How can we design a frustum bucket that holds exactly 50 L?'; and (4) place the new questions on the board and sort them into 'Answered', 'Partly answered' and 'Still to investigate'.	group. Find the questions we can now answer and put your evidence on a sticky note.' Give WAIT TIME of 10-15 seconds after asking: 'Which question on our board has changed the most today?' Cold-call 3 groups to read an answered question and its evidence. Ask: 'What new question came up for you? Who is curious about pyramids with the top cut off?' Select 2 new questions and read them aloud. Link to the next lesson: 'Next time we cut the top off a pyramid. Do you think the same method will work?' Make sure the question about designing a frustum for a required capacity stays visible as the driving question that the class will return to in Lesson 7. Praise respectful listening.	record of progress: sorting questions into 'Answered', 'Partly answered' and 'Still to investigate' makes learners aware of how their understanding is growing and gives them ownership of the investigation. Asking learners to generate new questions encourages curiosity and connects the lesson to Lesson 5.	match answers to the right DQB questions with correct evidence (58.52 L, formula, similar triangles); (b) new questions show transfer (for example, pyramids, designing a container of required capacity); (c) all group members take part in writing and placing notes. Record which groups produced a question about frustums of pyramids; they are ready for Lesson 5. Note learners who still ask 'why is the bucket not half of the drum?' for a quick follow-up.
Model Building Phase  VIDEO: Determining similar triangles <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Solve similar triangles (advanced) Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Learners update the single cumulative class model diagram. The diagram already shows cylinder-cone (3 : 1, Lesson 2) and cuboid-pyramid (3 : 1, Lesson 3). Today they add the frustum of a cone: a large cone ($r = 21$ cm, $h = 180$ cm, $83,160$ cm³) with a dashed small cone ($r = 14$ cm, $h = 120$ cm, $24,640$ cm³) removed, leaving the bucket ($R = 21$ cm, $r = 14$ cm, $h = 60$ cm, 58.52 L). They add the formula $V = \frac{1}{3} \pi h (R^2 + Rr + r^2)$, the label 'cover B 27.72 L < bucket C 58.52 L < drum A 83.16 L' and a note on the bucket being about 70% of the drum. Individually, they write 2-3 sentences in their notebooks to complete: 'The bucket holds _____ because _____.' Then they revise their Lesson 1 prediction in a different colour and write one thing that surprised them.	Display the model on the wall or chart paper. Say: 'Today we add the bucket. What do we draw first? Which part is removed?' Invite 3 volunteers to add the large cone, the dashed small cone and the labels. Give WAIT TIME of 10-15 seconds before asking: 'How does this part of the model connect to what we added in Lessons 2 and 3?' Cold-call 3 learners to link ideas (one-third idea, large minus small). Ask: 'Mama Chebet asks how to design a bucket that holds exactly 50 L. What do you already know that could help?' Cold-call 2 learners. Circulate during the writing and look for the key idea: shape, not just height and opening, determines volume. Collect notebooks or exit tickets for review. Close: 'Mr. Kiprop's bucket holds 58.52 L: more than the cover, less than the drum. Why?'	Cumulative modelling and prediction revision: adding the frustum to the single class model shows how the cone, pyramid and frustum ideas fit together and gives learners a visual tool to resolve the anchoring phenomenon. Revising Lesson 1 predictions in another colour makes the change in thinking visible and builds metacognition (Developing and Using Models).	Observable indicators: (a) model shows large cone, removed small cone and bucket with correct dimensions and volumes; (b) learners correctly explain that the bucket is between the cover and the drum because it is a large cone minus a small cone; (c) the revised prediction includes numbers and the formula; (d) learners link the model to Lessons 2 and 3. Exit ticket: 'Write the formula for the frustum of a cone, find the capacity of bucket C and explain in one sentence why it holds less than drum A.' Success criterion: 58.52 L and a sentence about shape or tapering.

D. TEACHER REFLECTION

After the lesson, consider: (1) Did learners measure the perpendicular height (60 cm) and not the slant height? Which groups needed correction? (2) How many learners found the missing height 120 cm using similar triangles, and who needed the ratio 2 : 3 explained again? (3) Could they explain why subtracting the small cone works? (4) How did learners respond to the algebraic derivation, and would a numerical pattern first have helped? (5) Did the 58.52 L result resolve the phenomenon for them, or do some still think the bucket should be about half the drum (it is about 70%)? (6) Which new DQB questions show readiness for Lesson 5? (7) Was WAIT TIME of 10-15 seconds used, and were quiet learners drawn in through cold-calls and pair talk? Plan reteaching using the paper cone model for learners who struggle.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	We measured bucket C: top radius $R = 21$ cm, bottom radius $r = 14$ cm and perpendicular height $h = 60$ cm. By extending the slanting sides we found a small cone ($h = 120$ cm) on top of the bucket and a large cone ($h = 180$ cm), using the ratio $r : R = 2 : 3$. Large cone = $83,160 \text{ cm}^3$, small cone = $24,640 \text{ cm}^3$.
What did I learn?	The volume of a frustum of a cone is $V(\text{large cone}) - V(\text{small cone}) = \frac{1}{3} \pi h (R^2 + Rr + r^2)$. Similar triangles give the missing height of the small cone. Using $\pi = 22/7$, bucket C holds $58,520 \text{ cm}^3 = 58.52 \text{ L}$.
How does this explain the phenomenon?	The cover B (27.72 L), bucket C (58.52 L) and drum A (83.16 L) are all about 60 cm tall with a 42 cm opening, but they hold different amounts because of shape. The cover tapers to a point, the bucket tapers only to a 28 cm base, and the drum does not taper. So the bucket holds more than the cover and less than the drum (about 70% of the drum). The same large-minus-small reasoning can be used to design a frustum container of any required capacity.

LESSON 5 (80 minutes (2 x 40-minute lessons)): Volume of a Frustum of a Pyramid: Designing the Maize Bin

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners extend the large-minus-small method (Lesson 4) and the pyramid formula (Lesson 3) to find the volume of a frustum of a pyramid, $V = \frac{1}{3} h (A_1 + A_2 + \sqrt{A_1 A_2})$, and use maize bulk density to convert capacity to mass. This advances the driving question: how can we use mathematics to design a frustum-shaped container that holds exactly the capacity we need?
Knowledge	By the end of the lesson the learner should be able to: (a) calculate the volume and capacity of a frustum of a pyramid using $V = \frac{1}{3} h (A_1 + A_2 + \sqrt{A_1 A_2})$; (b) verify the result by subtracting the small pyramid from the large pyramid; (c) convert cm^3 to litres ($1000 \text{ cm}^3 = 1 \text{ L}$) and litres to kilograms of maize using 0.75 kg per litre; (d) apply volume and capacity in real-life situations such as grain storage in the Rift Valley; (e) appreciate volumes and capacities in real-life situations.
Skills	Using Mathematical Reasoning and Computational Thinking (NGSS SEP 5): calculating areas, a square root, volume, capacity and mass. Developing and Using Models (SEP 2): extending the class cumulative model to the frustum of a pyramid. Constructing Explanations (SEP 6): explaining why the frustum formula has three terms. Communication and collaboration in groups.
Attitudes	Critical thinking and problem solving: the learner shows open-mindedness and creativity when discussing with peers and working out the volume and capacity of containers. Unity: tolerance of different opinions when working with peers. Respect: listening to others' opinions when discussing appropriate formulas. Environment and technology: using appropriate containers from the local environment to work out capacity.
Key Inquiry Question	Why is the knowledge of volume and capacity useful?
Purpose in Storyline	Lesson 4 resolved the bucket (C): a frustum of a cone holds 58.52 L, between the cover (B, 27.72 L) and the drum (A, 83.16 L). Lesson 5 shows that the same reasoning works for pyramids by building on the pyramid formula from Lesson 3. Learners move from explaining existing containers toward designing one: a square-based maize bin model for the Rift Valley that holds a required capacity and a known mass of maize. This prepares learners for the mixed word problems in Lesson 6.
Safety Notes	Use only dry maize or sand for filling. Wipe up spills at once to avoid slipping. Handle card or plastic models carefully; use blunt scissors; keep bags of maize off the floor walkways. Calculators and rulers are handled with care and shared fairly.

B. LESSON OVERVIEW

Ms. Wanjiru shows a card model of a maize bin for farmers in the Rift Valley: a square-based frustum with top side 60 cm, bottom side 30 cm and height 40 cm. She reminds the class of Mama Chebet's bucket (C) and asks: if the bucket is a large cone minus a small cone, can the bin be a large pyramid minus a small pyramid? Learners predict the capacity and the mass of maize it holds. They then complete the large-minus-small pyramid method ($96,000 \text{ cm}^3 - 12,000 \text{ cm}^3 = 84,000 \text{ cm}^3$), compare it with $V = \frac{1}{3} h (A_1 + A_2 + \sqrt{A_1 A_2})$ using $A_1 = 3,600 \text{ cm}^2$, $A_2 = 900 \text{ cm}^2$ and $\sqrt{A_1 A_2} = 1,800 \text{ cm}^2$, and confirm $84,000 \text{ cm}^3 = 84 \text{ L}$. Using the maize bulk density of 0.75 kg per litre, they find the bin holds 63 kg of maize. The class updates the DQB and extends its cumulative model so that it covers the cylinder, cone, pyramid, cone frustum and pyramid frustum.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Volume of a Pyramid - Overview Source: CK-12  Search ARES for similar videos  EXERCISE: Volume of prisms and pyramids Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Ms. Wanjiru holds up a card model of a square-based maize bin: top side 60 cm, bottom side 30 cm, height 40 cm, and labels these on the board. She recalls Mr. Kiprop's question about why the bucket (C) is not as big as the drum (A). Learners work individually for 3 minutes in their notebooks to answer: (1) Will the bin hold more or less than a square prism of the same height and top side ($60 \times 60 \times 40 = 144,000 \text{ cm}^3$)? (2) Is the frustum of a pyramid the 'large pyramid minus small pyramid', as the bucket was 'large cone minus small cone'? (3) Estimate the capacity in litres and the mass of maize in kg. They then share with a partner and record predictions on a class chart. The predictions stay on the board until the Explain Phase.	Say: 'Last lesson we explained Mama Chebet's bucket by taking a small cone away from a large cone. Today a farmer in the Rift Valley wants a maize bin shaped like this. Before I teach anything, write your prediction.' Give WAIT TIME of 10-15 seconds after each question before anyone speaks. Cold-call 4 learners for their litre estimates without commenting on right or wrong; write all four on the board. Ask: 'Who thinks the bin holds more than the 83.16 L drum? Why?' Cold-call 2 learners to give reasoning. Do NOT confirm any number. Circulate and note learners who reason from the cone-frustum idea.	Predict-Observe-Explain (POE) with Making Thinking Visible: writing and sharing predictions before any calculation exposes prior ideas (such as 'bigger top means bigger volume' or 'a frustum is half the prism'), so learners have something to test against evidence. Connecting to the phenomenon through the bucket draws on Lesson 4 knowledge.	Observe: Do learners use the idea of subtracting a small solid from a large one? Do they distinguish volume (cm^3) from capacity (L)? Indicator: at least 20 of 30 learners write a numerical prediction with units; note the range of estimates and learners who mention a missing small pyramid. Misconception check: learners who simply average the two base areas and multiply by the height.
Observe Phase  VIDEO: Volume of pyramids intuition Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Volume of Pyramids (Flexbook) Source: CK-12  Search ARES for similar readings	Part 1 (large minus small): In groups of 4, learners receive a diagram of the frustum extended to a full pyramid. They are told the large pyramid has base side 60 cm and the small removed pyramid has base side 30 cm. Using the 2:1 ratio of sides (as the 3:2 ratio helped in Lesson 4) they find the missing height 40 cm, so the large pyramid has height 80 cm. They calculate: large pyramid = $\frac{1}{3} \times 3,600 \times 80 = 96,000 \text{ cm}^3$; small pyramid = $\frac{1}{3} \times 900 \times 40 = 12,000 \text{ cm}^3$; frustum = $96,000 - 12,000 = 84,000 \text{ cm}^3$. Part 2 (model fill): Using a card frustum model (top 60 cm, bottom 30 cm, height 40 cm) or a pyramid-shaped scaled bin where available, groups observe how the small pyramid (the 'Lesson 3' grain-store shape,	Distribute the diagrams and evidence tables. Say: 'Use the method from Lesson 4. What is the ratio of the side of the small pyramid to the side of the large one? How does that help you find the missing height?' Give WAIT TIME of 10-15 seconds before taking answers. Cold-call 3 groups to state the height of the large pyramid and justify it. Circulate: ask each group 'Which height are you using, the vertical height or the slant edge?' (h is always vertical). Check that learners find $\sqrt{3,600 \times 900}$ correctly, noting that it also equals $60 \times 30 = 1,800$. Prompt: 'Is 1,800 between 900 and 3,600? Why does that make sense?' Do not state the formula before groups have computed both methods; introduce the name of	Evidence Table and Two-Method Triangulation: recording the large-minus-small result next to the formula result lets learners see that two methods agree, building trust in the formula. Pattern-noticing: comparing with the cone frustum formula helps them see the three-term structure ($A_1 + A_2 + \sqrt{A_1 A_2}$ mirrors $R^2 + Rr + r^2$).	Observable indicators: groups obtain $96,000 \text{ cm}^3$, $12,000 \text{ cm}^3$ and $84,000 \text{ cm}^3$ correctly; groups obtain $\sqrt{A_1 A_2} = 1,800 \text{ cm}^2$; table has units on every line. Check: at least 6 of 8 groups reach $84,000 \text{ cm}^3$ by both methods. Common errors to record: using $h = 40$ for the large pyramid (should be 80), forgetting the $\frac{1}{3}$, using the sum of the sides instead of the areas.

	side 30 cm, h 40 cm) fits at the top of the removed part. If no big model is available, groups use the teacher's demonstration and the diagram. Part 3 (formula test): Groups compute $A_1 = 60 \times 60 = 3,600 \text{ cm}^2$, $A_2 = 30 \times 30 = 900 \text{ cm}^2$, $A_1 \times A_2 = 3,240,000$, $\text{sqrt} = 1,800 \text{ cm}^2$, then $V = 1/3 \times 40 \times (3,600 + 900 + 1,800) = 1/3 \times 40 \times 6,300 = 84,000 \text{ cm}^3$. They record both results in an evidence table.	the formula only after they notice that both give $84,000 \text{ cm}^3$.		
Explain Phase  VIDEO: Volume of a Pyramid - Overview Source: CK-12  Search ARES for similar videos  EXERCISE: Volume of Pyramids Practice Source: CK-12  Search ARES for similar readings	Whole-class discussion followed by individual work. (1) Learners compare their Predict Phase estimates with $84,000 \text{ cm}^3$ and explain why the answer is less than the prism ($144,000 \text{ cm}^3$) but more than the small pyramid. (2) Learners convert: $84,000 \text{ cm}^3 / 1,000 = 84 \text{ L}$. They compare with the drum (83.16 L): the bin holds slightly more than the drum, even though the drum is a plain cylinder. (3) Mass of maize: Ms. Wanjiru introduces bulk density: 1 litre of maize has a mass of about 0.75 kg. Learners compute $84 \times 0.75 = 63 \text{ kg}$. They discuss why a farmer in the Rift Valley would want to know this (how many 90 kg bags, planning storage, selling by weight). (4) Individual task: 'A new bin has top side 40 cm, bottom side 20 cm and height 30 cm. Find A_1, A_2, $\text{sqrt}(A_1A_2)$, the volume, the capacity and the mass of maize.' Answer: $A_1 = 1,600$, $A_2 = 400$, $\text{sqrt} = 800$, $V = 1/3 \times 30 \times 2,800 = 28,000 \text{ cm}^3 = 28 \text{ L} = 21 \text{ kg}$. (5) Extension for the design question: 'What bottom side would give the same height and top side (60 cm, h 40) but a smaller capacity?' Learners discuss how changing A_2 changes the volume.	Say: 'Compare your prediction with the evidence. What did you get right? What did you change your mind about?' Give WAIT TIME of 10-15 seconds, then cold-call 4 learners. Ask: 'Why do you think the formula has three terms and not just the two areas?' Wait 10-15 seconds; cold-call 3 learners and record their reasoning on the board; connect to the cone frustum where R^2, Rr and r^2 appear and $\text{sqrt}(A_1A_2)$ plays the role of Rr. Ask: 'If the top and bottom were equal, what shape would we have? What does the formula give?' ($A_1 = A_2 = A$, $V = 1/3 h \times 3A = Ah$: a prism; use this as a check and as a link to Lesson 3's cuboid.) Ask: 'If A_2 were 0, what solid is it?' (a pyramid; $V = 1/3 h A_1$.) Then introduce density: 'Capacity tells us litres. To tell the farmer how heavy the maize is, we need mass per litre.' Have 5 learners show the working for the new bin on the board and check as a class. Do not re-teach the cone-and-cylinder ratio (Lesson 2) or the cone frustum derivation (Lesson 4); refer to them briefly.	Claim-Evidence-Reasoning (CER) with Special-Case Checking: learners claim a formula is correct, support it with $84,000 \text{ cm}^3$ obtained two ways, and reason through limiting cases ($A_1 = A_2$ gives a prism; $A_2 = 0$ gives a pyramid). Unit chain reasoning (cm^3 to L to kg) shows why conversions matter for real farmers' decisions.	Exit-style check on the 40-20-30 bin: indicators are $A_1 = 1,600 \text{ cm}^2$, $A_2 = 400 \text{ cm}^2$, $\text{sqrt} = 800 \text{ cm}^2$, $V = 28,000 \text{ cm}^3$, 28 L, 21 kg. At least 22 of 30 learners should get all six values; note anyone who uses 0.75 as L per kg instead of kg per L. Listen for explanations of the three terms and of the prism and pyramid limiting cases. Provide a quick re-teach with the evidence table for learners with two or more errors.

Driving Question Board (DQB) Creation  VIDEO: Volume of pyramids intuition Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Volume of Pyramids: Fluctuating Height (PLIX) Source: CK-12  Search ARES for similar readings	The DQB is on the classroom wall. Each group writes one new question or an answer on sticky notes, for example: 'Can we find a frustum bin that holds exactly 100 L?', 'Does the formula work if the base is a rectangle?', 'Why does the square root of $A1 \times A2$ appear?', 'How many 2 kg tins of maize fit in the bin?'. Groups move earlier questions (for example 'Why does the cylinder hold three times the cover?' and 'Why does the bucket hold less than the drum?') to the Answered column where evidence supports it, and add this lesson's evidence: bin 84 L and 63 kg. Learners also add a note on what they still need for the design question: measuring and building real containers (Lesson 7) and solving mixed problems (Lesson 6).	Say: 'Look at the board. Which of our questions can we now answer with evidence? Which new questions did today raise?' Give WAIT TIME of 10-15 seconds before groups write. Cold-call 3 groups to read one note aloud and explain the evidence. Ask: 'Which question will help us design a container for a farmer?' Facilitate 2 learners to move notes to Answered. Highlight at least one question about a rectangular base and say it will be tested in Lesson 6 problems with different dimensions. Keep the DQB visible for Lesson 6.	Driving Question Board as a Learning Tracker: sorting questions into answered and open keeps the unit goal visible and models how scientists and mathematicians refine questions as evidence grows. Learners link each answered question to specific evidence.	Indicators: every group posts at least one new question or evidence note; at least 3 questions refer to design (capacity needed, dimensions, mass); at least 2 learners correctly move a question to Answered with the evidence (84,000 cm³, 63 kg, 58.52 L bucket). Note vague questions and ask groups to sharpen them with numbers and units.
Model Building Phase  VIDEO: Metric system: units of volume <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Volume Of A Pyramid Source: math_expression  Search ARES for similar readings	The class has a single cumulative diagram on chart paper that grew in earlier lessons (cylinder and cone relationship, pyramid, cone frustum). Groups add the pyramid frustum: a square-based frustum drawn inside its large pyramid with the small pyramid removed, labelled top side 60 cm, bottom side 30 cm, h 40 cm, large pyramid 96,000 cm³, small pyramid 12,000 cm³, frustum 84,000 cm³ = 84 L = 63 kg maize. Beside it they write the formula $V = \frac{1}{3} h (A1 + A2 + \sqrt{A1 A2})$ and draw an arrow linking it to the cone frustum formula $V = \frac{1}{3} \pi h (R^2 + Rr + r^2)$ to show the shared 'large minus small' idea. Each learner then writes a short summary in their notebook: one sentence explaining how a frustum of a pyramid relates to a pyramid, and one sentence explaining how to	Say: 'Our model must now explain two kinds of frustum. Where does the pyramid frustum sit on our diagram?' Give WAIT TIME of 10-15 seconds. Cold-call 3 groups to explain one part of the added diagram. Check that h is vertical and that units appear on every label. Ask: 'What is the same about the cone frustum and the pyramid frustum? What is different?' Wait 10-15 seconds and cold-call 2 learners. Ask: 'If a farmer wanted exactly 100 L, which measure would you change first, and how would you check?' Accept reasoned suggestions (for example, increase the height or the top side) and note that they will test designs in Lesson 7. Collect notebooks for review. Do not introduce new formulas here.	Cumulative Model Revision: adding the frustum of a pyramid to a single class diagram makes the connections among cylinder, cone, pyramid and frustum explicit. Learners use the model to reason about design changes and to predict outcomes, which is the purpose of modelling.	Indicators: the revised diagram has correct labels and units; the formula is linked to the large-minus-small method; learner summaries mention 'one third', 'large minus small' and 'multiply litres by 0.75'. At least 24 of 30 learners write both sentences correctly. Look for sound design reasoning on the 100 L question (for example, 'make the height greater' or 'increase A1 or A2').

	turn litres into kilograms of maize. They also complete the summary table (what we observed, what we learned, how we explain it) and answer: 'How would you change the bin so that it holds exactly 100 L?'			
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D. TEACHER REFLECTION

Which learners confused the vertical height with the slant edge or forgot the missing small pyramid? Did the two-method comparison (96,000 - 12,000 and the formula) convince learners, or did some still treat the formula as a rule to memorise? Did the limiting cases ($A_1 = A_2$ and $A_2 = 0$) help learners see why the formula has three terms? Were the unit chain (cm^3 to L to kg) and the density idea clear, or did some multiply when they should divide? Which DQB questions will you carry into Lesson 6 problem solving, and which learners need extra practice with square roots of areas before they work on mixed word problems?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	We observed that a square-based maize bin model with top side 60 cm, bottom side 30 cm and height 40 cm gives 84,000 cm ³ both by subtracting the small pyramid (12,000 cm ³) from the large pyramid (96,000 cm ³) and by the formula with $A_1 = 3,600 \text{ cm}^2$, $A_2 = 900 \text{ cm}^2$ and $\text{sqrt}(A_1 A_2) = 1,800 \text{ cm}^2$.
What did I learn?	We learned that the volume of a frustum of a pyramid is $V = \frac{1}{3} h (A_1 + A_2 + \text{sqrt}(A_1 A_2))$, that $84,000 \text{ cm}^3 = 84 \text{ L}$, and that maize has a bulk density of 0.75 kg per litre, so the bin holds $84 \times 0.75 = 63 \text{ kg}$ of maize.
How does this explain the phenomenon?	We can explain that, like Mama Chebet's bucket (C) as a large cone minus a small cone, a pyramid frustum is a large pyramid minus a small pyramid, which is why its volume lies between those of the small and large pyramids; we can use this to design a frustum container of the capacity a farmer needs.

LESSON 6 (80 minutes (2 x 40-minute lessons)): Problem Solving with Real Containers: Turkana Tank, Githeri Pot and Frustum Cups

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners choose the correct volume formula and unit conversion for multi-step Kenyan word problems, and use division of one capacity by another, to show that the formulas from Lessons 2-5 can predict and design real container capacities such as the Kericho drum (A), cover (B) and bucket (C).
Knowledge	By the end of the lesson the learner should be able to: (a) determine the volume and capacity of cones and pyramids; (b) compute volume and capacity of a frustum of a cone; (d) apply volume and capacity in real-life situations; (e) appreciate volumes and capacities in real life situations. Specific facts: Turkana tank (cone, $r = 1.4 \text{ m}$, $h = 3 \text{ m}$) = $6.16 \text{ m}^3 = 6,160 \text{ L}$; githeri pot (frustum, $R = 30 \text{ cm}$, $r = 24 \text{ cm}$, $h = 35 \text{ cm}$) = 80.52 L , enough for 100 people at 0.5 L each (50 L); frustum cup ($R = 6 \text{ cm}$, $r = 3 \text{ cm}$, $h = 7 \text{ cm}$) = 462 cm^3 , so drum A ($83,160 \text{ cm}^3$) holds 180 cups.
Skills	Mathematical reasoning and problem solving: identify the solid, select the matching formula, substitute with $\pi = 22/7$, convert units (cm^3 to L , m^3 to L), and check the answer for sense. Communicate and justify formula choices in a group presentation. Divide one capacity by another to find a number of containers.
Attitudes	Critical Thinking and Problem Solving: open-mindedness and creativity when discussing with peers and working out volume and capacity of the containers. Respect: the learner listens to others' opinions while discussing appropriate formulas. Unity: tolerance of different opinions while working with peers. Appreciation of the usefulness of volume and capacity in real life.
Key Inquiry Question	Why is the knowledge of volume and capacity useful?
Purpose in Storyline	Lessons 2-5 gave learners four formulas: cone (Lesson 2), pyramid (Lesson 3), frustum of a cone (Lesson 4) and frustum of a pyramid (Lesson 5). Lesson 1 gave the unit conversions. Lesson 6 moves from the Kericho containers to new Kenyan contexts, so learners must decide which formula and which conversion to use. This answers the second half of the driving question: using mathematics to design a frustum-shaped container with the capacity we need. The DQB is updated with consolidated understanding, and learners prepare for the Lesson 7 modelling project.
Safety Notes	Pour water only over trays or buckets and wipe spills immediately to prevent slipping. Calculators are handled with care and shared fairly. Learners lift the drum and bucket models in pairs and do not stand on desks to display charts. Keep walkways clear during gallery presentations.

B. LESSON OVERVIEW

Mama Chebet's neighbour, Mr. Kiprop, asked why his bucket is not as big as the drum. Lessons 2-5 answered that using formulas. Now Ms. Wanjiru challenges the class: can the same mathematics solve problems elsewhere in Kenya? Groups of four tackle three problems: (1) sizing a conical water tank for a Turkana homestead, (2) checking whether a frustum githeri pot can cook for 100 people at a community meal in Nairobi, and (3) finding how many frustum cups fill the Kericho cylindrical drum A. Learners first predict which formula each problem needs and the rough size of the answers. They then work through the problems with a decision routine (Name the solid, Pick the formula, Convert units, Check for sense). Groups present and defend their choices, and the class updates the DQB and the cumulative class model. The lesson applies the drum (A), cover (B) and bucket (C) reasoning to new contexts rather than re-teaching the formulas.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Volume word problem: water tank Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Volume of cylinders, spheres, and cones word problems Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Ms. Wanjiru shows three problem cards without numbers: a Turkana homestead needs a conical water tank; a Nairobi community meal needs a frustum githeri pot for 100 people; and Mama Chebet wants to know how many small frustum cups fill her drum A. Individually, learners write on a sticky note: (1) the solid shape in each problem; (2) the formula they think they will need, chosen from the four they learned in Lessons 2-5; (3) one unit conversion they expect to need. Then learners estimate in pairs: Will the pot hold 50 L? Will the cup count be nearer 18, 180 or 1,800? Learners link back to the Kericho containers: the cup looks like a tiny version of bucket C, so the same frustum formula should apply.	Display the three cards and say: 'Before we calculate anything, I want your thinking. What solid is each container, and why?' Give 1 minute silent writing, then WAIT TIME of 10-15 seconds after each question before taking answers. Cold-call 4 learners for the Turkana tank, githeri pot and cup shapes, and 3 more for their formula choices; ask each 'What in the shape told you that?' Ask: 'Which unit conversion from Lesson 1 will we need? Cubic metres to litres? Cubic centimetres to litres?' Do not reveal any answers. Record every estimate on the board as 'Prediction Bank' with initials. Say: 'Disagreement is welcome. We will test these predictions with calculation.'	Prediction with Written Commitment: writing the formula choice and an estimate before calculating makes prior ideas visible, so learners can later confront their own reasoning. Revising wrong predictions becomes evidence of learning, not failure.	Observe whether learners identify the Turkana tank as a cone, the githeri pot and cup as frustums of cones, and the drum A as a cylinder. Indicators: 80% of sticky notes name a solid and a matching formula; learners spot that the cup (frustum) and bucket C share the same shape. Note learners who pick the pyramid formula for round containers, or who forget that m^3 must be converted to litres, and give them targeted prompts during group work.
Observe Phase  VIDEO: Challenging Problems Involving Capacity in Liters - Word Problems Source: CK-12  Search ARES for similar videos  EXERCISE: Apply volume of solids Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Groups of four receive the Problem Set with the data. Problem 1: Turkana homestead conical tank, radius 1.4 m, height 3 m; find the volume in m^3 and the capacity in litres. Problem 2: githeri frustum pot, top radius 30 cm, bottom radius 24 cm, height 35 cm; find the capacity in litres and decide whether it holds the 50 L needed for 100 people at 0.5 L each. Problem 3: frustum cup, top radius 6 cm, bottom radius 3 cm, height 7 cm; find its volume in cm^3, then divide the volume of drum A ($83,160\text{ cm}^3$) by it to find how many cups fill the drum. Each learner has a role (reader, calculator, recorder, checker) and rotates roles between problems. Learners follow the four-step	Circulate with a checklist. Ask scaffolding questions rather than giving answers: 'Which solid is this? How do you know?'; 'What does h mean here: vertical height or slant height?'; 'Your answer is 6.16. In what unit? What does the Turkana family actually need to know, cubic metres or litres?' For groups stuck on Problem 3, ask: 'What is the question really asking: a volume or a count? What operation gives a count?' Stop the class twice for 'checker pause': 'Checkers, ask your group to explain one step in their own words.' Cold-call 3 checkers per pause. WAIT TIME of 10-15 seconds after each question. If a group finds $80,520\text{ cm}^3$ but writes 80.52 mL, ask: 'Is 1 L more or less	Structured Problem-Solving Routine with Formula Selection Table: the four-step routine (Name, Pick, Convert, Check) slows learners down so that they connect shape to formula before computing. The table makes the reasoning visible, so errors in formula choice and units can be traced and corrected.	Observable indicators: each group correctly identifies cone for the tank and frustum of a cone for the pot and cup; substitutes R, r, h correctly; converts 6.16 m^3 to 6,160 L and $80,520\text{ cm}^3$ to 80.52 L; compares 80.52 L with 50 L; divides 83,160 by 462 to get 180. Tally on a checklist: correct formula (Y/N), correct conversion (Y/N), sense-check written (Y/N). Common errors to watch: using slant height; forgetting $1/3$; squaring incorrectly (e.g. $30 \times 24 = 720$ mistaken); dividing 462 by 83,160; mixing m^3 and L.

	routine: Name the solid, Pick the formula, Convert units, Check for sense. They work with $\pi = 22/7$. Evidence they collect: Problem 1: $V = 1/3 \times (22/7) \times 1.96 \times 3 = 6.16 \text{ m}^3 = 6,160 \text{ L}$. Problem 2: $R^2 + Rr + r^2 = 900 + 720 + 576 = 2,196$, so $V = 1/3 \times (22/7) \times 35 \times 2,196 = 80,520 \text{ cm}^3 = 80.52 \text{ L}$, which is greater than 50 L, so the pot fits the meal with 30.52 L to spare. Problem 3: $V = 1/3 \times (22/7) \times 7 \times 63 = 462 \text{ cm}^3$, and $83,160 / 462 = 180$ cups. Groups fill in a Formula Selection Table recording solid, formula, working, units and sense-check.	than $1,000 \text{ cm}^3$? Look at your Lesson 1 conversion chart.' Have an early-finisher extension ready: 'If the pot must serve 150 people at 0.5 L each, does it still work?' (75 L, which is less than 80.52 L, so it still fits.)		
Explain Phase  VIDEO: Word problems with volume Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Solid geometry word problems Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Each group prepares a 2-minute justification on a chart for one problem and presents it in a gallery-style 'Share and Defend' session. The presenting group must state: (1) which solid and why, (2) which formula and why not the others, (3) the unit conversion and why it is needed, (4) a sense-check. The audience asks one question each. Learners explain connections to the phenomenon: the frustum cup uses the same large-cone-minus-small-cone idea as bucket C (58.52 L); the Turkana cone tank uses the same one-third idea that makes cover B 27.72 L, one third of drum A at 83.16 L. They also explain why the ratio of volumes (drum A to cup) gives a count of 180 and what that tells Mama Chebet: she could serve 180 cups from one full drum. Learners compare the pot answer, 80.52 L, with the need of 50 L and write a decision statement: 'The pot is large enough because...'. Where groups differ (for example rounding), the class resolves the	Say: 'Convince me, and convince your classmates. I am listening for a reason, not only an answer.' Assign problems so each is presented by at least one group. After each presentation, ask the audience: 'Who can restate the group's reasoning in your own words?' and cold-call 3 learners with WAIT TIME of 10-15 seconds. Probe: 'Why would a pyramid formula have been wrong for the cup?'; 'The cup is a frustum. Where is the small cone that was removed? How tall would it be?' (Cup: small cone height 7 cm, because the radii are 6 and 3, a ratio of 2:1, so the full cone is 14 cm tall and the missing small cone is 7 cm.) Connect back: 'How is the cup like Mr. Kiprop's bucket?' Ask: 'How do the formulas help a designer? If you needed a pot to hold exactly 100 L, what would you change?' Affirm correct reasoning, and correct units publicly but kindly. Reinforce respect: 'Thank the group that challenged you.'	Claim-Evidence-Reasoning Gallery with Peer Restating: presenters make a claim (the answer), provide evidence (working and units), and give reasoning (why this formula). Peer restating checks that the audience understands and builds the habit of mathematical justification.	Observable indicators: presenters give a reason for formula choice using shape features (circular base, two parallel circular faces, tapered to a point) rather than 'because the teacher said so'; correctly state 6,160 L, 80.52 L (fits 50 L) and 180 cups; audience members restate accurately. Use a 3-point scale on a rubric: 3 = correct formula, conversion and sense-check justified; 2 = correct answer with weak justification; 1 = answer with formula or unit errors. Note learners who still confuse volume (cm^3) with capacity (L) for follow-up.

	difference by returning to the stated constants.			
Driving Question Board (DQB) Creation  VIDEO: Volume word problem: water tank Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Volume of cylinders, spheres, and cones word problems Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	The DQB is posted on the wall. Learners revisit earlier questions, such as 'Why does the drum hold three times the cover?' and 'Why does the bucket hold less than the drum?', and sort them into 'Answered with evidence', 'Partly answered' and 'New questions'. Each group moves two sticky notes and adds one new question in response to today's problems, for example 'How would we design a frustum pot of exactly 100 L?' or 'Does the same method work for a pyramid-frustum bin?' They write a consolidated answer card: 'We can use mathematics to find the capacity of any cone, pyramid or frustum container because...' and attach it to the DQB with the evidence of 6,160 L, 80.52 L and 180 cups.	Say: 'Walk to the board. Which questions can we now answer with evidence? Show me where the evidence is.' Give WAIT TIME of 10-15 seconds, then cold-call 4 groups to justify moving a question. Ask: 'What does today add to our driving question: why are the containers different, and how can we design a frustum container for the capacity we need?' Highlight questions that point to Lesson 7: 'We will test this by making our own models and measuring. Which new question will our models help us answer?' Make sure questions about error and accuracy are noted to be tackled in Lesson 7. Do not answer the new questions; mark them for the project.	Driving Question Board Sorting: moving questions between categories shows learners how evidence has shifted their understanding, and keeps the unit coherent by linking each lesson's evidence to the driving question.	Observable indicators: learners move questions with a reason that cites specific evidence (for example, 'we showed $83,160 / 462 = 180$ '); new questions are specific and mathematical (design, accuracy, other shapes) rather than vague; every group contributes at least one note. Check that answer cards use correct vocabulary (volume, capacity, frustum) and units.
Model Building Phase  VIDEO: Challenging Problems Involving Capacity in Liters - Word Problems Source: CK-12  Search ARES for similar videos  EXERCISE: Apply volume of solids Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Learners update the single cumulative class model diagram, which already shows drum A (83.16 L), cover B (27.72 L), bucket C (58.52 L as large cone $83,160 \text{ cm}^3$ minus small cone $24,640 \text{ cm}^3$), the pyramid and the maize bin. In groups they add a new panel titled 'Choosing a Formula' with a decision flowchart: Is the container round or square-based? Does it taper to a point (cone/pyramid) or is it cut off (frustum)? Then they add the formulas and a 'Convert Units' strip ($1 \text{ cm}^3 = 1 \text{ mL}$; $1000 \text{ cm}^3 = 1 \text{ L}$; $1 \text{ m}^3 = 1000 \text{ L}$). They also add three worked examples: Turkana tank 6,160 L, githeri pot 80.52 L (fits 50 L) and cup 462 cm^3 giving 180 cups per drum A. Each learner	Say: 'Our model must now help someone who has never met our containers choose the right formula. What must go on the flowchart?' Give WAIT TIME of 10-15 seconds, then cold-call 3 groups for one branch of the flowchart each. Ask: 'Where on our diagram does the 3:1 relationship show up in today's problems?' (Turkana cone tank uses the one-third factor.) 'Where does the large-minus-small idea show up?' (frustum pot and cup). Collect exit responses (about 5 minutes). Preview Lesson 7: 'Next time you will design and build your own container, calculate its volume, and test it with water or sand. What errors might you expect?' Do not give the expected error range	Cumulative Model Revision with Decision Flowchart: turning formula choice into a visual flowchart consolidates the relationships among cylinder, cone, pyramid and frustum, and lets learners use the model as a predictive tool for design.	Observable indicators: the flowchart sorts solids correctly by base shape and by whether they are cut off; all four formulas are written with correct symbols (r, R, h, A1, A2) and h noted as vertical height; conversions are included; worked examples have correct values. Exit responses should name a formula that matches the container and mention a conversion. Record learners who confuse round and square bases or omit units, and revisit them in Lesson 7 groups.

	writes a short individual exit response: 'Which formula would you use to design a frustum container for a given capacity, and what would you need to decide first?'	yet.		
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D. TEACHER REFLECTION

Which groups chose formulas from the shape of the container and which relied on guessing or copying? Did any learners mix up volume (cm^3 , m^3) and capacity (L), or use slant height instead of vertical height? How well did the Formula Selection Table and four-step routine expose errors? Did the gallery presentations show real reasoning or only answers? Which DQB questions are still unanswered, and do they point clearly to the Lesson 7 modelling project? Which learners need extra support with dividing capacities ($83,160 / 462$) or with converting m^3 to litres before the project?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	We observed that three new Kenyan problems (a conical Turkana tank, a frustum githeri pot and a frustum cup) can each be solved by identifying the solid and applying the matching formula: the Turkana tank has $V = 6.16 \text{ m}^3 = 6,160 \text{ L}$, the githeri pot has $V = 80,520 \text{ cm}^3 = 80.52 \text{ L}$ (enough for 100 people at 0.5 L each, which needs 50 L), and each cup has $V = 462 \text{ cm}^3$.
What did I learn?	We learned to choose the correct formula from the shape of the solid, to convert units (cm^3 to L, m^3 to L) before answering, to check answers for sense, and to divide one capacity by another to find a count: drum A holds $83,160 / 462 = 180$ cups.
How does this explain the phenomenon?	We can now explain that the same mathematics that makes drum A (83.16 L) hold three times cover B (27.72 L) and bucket C hold 58.52 L also lets us size tanks, pots and cups, and design a frustum container of any capacity we need, because each container's capacity depends on its shape, its dimensions and the formula that matches it.

LESSON 7 (80 minutes (2 x 40-minute lessons)): Project: Model Making and Measurement – Building and Testing Our Own Kericho-Style Containers

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners design and build model containers shaped as a cone, a frustum of a cone and a frustum of a pyramid from locally available materials. They calculate each model's theoretical volume, check it by filling with water or sand, and explain the difference using percentage error. This tests whether the formulas that explain Mama Chebet's drum, cover and bucket really work, and it moves learners toward designing a frustum container of a required capacity.
Knowledge	By the end of the lesson, the learner should be able to: a) determine the volume and capacity of cones and pyramids; b) compute the volume and capacity of a frustum of a cone; c) calculate the volume and capacity of a frustum of a pyramid; d) apply volume and capacity in real-life situations; e) appreciate volumes and capacities in real-life situations. Formulas used ($\pi = 22/7$): cone $V = \frac{1}{3} \pi r^2 h$; frustum of a cone $V = \frac{1}{3} \pi h (R^2 + Rr + r^2)$; frustum of a pyramid $V = \frac{1}{3} h (A_1 + A_2 + \sqrt{A_1 A_2})$; $1 \text{ cm}^3 = 1 \text{ mL}$; $1000 \text{ cm}^3 = 1 \text{ L}$.
Skills	Planning and designing a model; measuring lengths accurately with a ruler; choosing the correct formula and substituting values; converting cm^3 to mL and L; carrying out a fair test by filling with water or sand; calculating percentage error = $(\text{measured} - \text{calculated}) / \text{calculated} \times 100$; identifying sources of error; documenting a project with photos or video clips. NGSS-aligned practices: Developing and Using Models; Planning and Carrying Out Investigations; Analyzing and Interpreting Data; Using Mathematics and Computational Thinking; Engaging in Argument from Evidence.
Attitudes	Critical Thinking and Problem Solving: the learner acquires open-mindedness and creativity skills when discussing with peers and working out the volume and capacity of the containers. Creativity and Imagination: the learner develops connection or networking skills when working out the volumes of frustums from cones and pyramids. Unity: the learner displays tolerance of different opinions while working with peers when making models of frustums as a learning resource. Respect: the learner listens to others' opinions while discussing appropriate formulas for working out volume and capacity of containers. PCI (Environment and technology): the learner uses appropriate containers from the local environment to work out the capacity of cones, pyramids, and their frustum.
Key Inquiry Question	Why is the knowledge of volume and capacity useful?
Purpose in Storyline	Lessons 2 to 6 gave learners the formulas for the cone, pyramid and both frustums, and practice using them on Kenyan problems. In Lesson 7 learners stop being only users of the formulas and become designers and testers. They build their own versions of the Kericho containers, predict capacity by calculation, and check it with water or sand. Seeing that the maths predicts what really happens, to within a small, explainable error, gives the evidence needed for Lesson 8, where the class explains why the drum, cover and bucket hold such different amounts.
Safety Notes	Cut cardboard with scissors on a flat desk, cutting away from the body; a teacher or a trusted learner handles any craft knife. Wipe up spilled water at once to prevent slipping. Use a tray or basin when filling models. Use clean, dry sand and avoid raising dust (wet the sand slightly or pour slowly). Do not use sharp-edged tin cans unless the rims are flattened. Wash hands after the activity and put all offcuts in the recycling box. Learners use phones or cameras only for project photos and videos, with the teacher's permission.

B. LESSON OVERVIEW

Ms. Wanjiru opens by reminding the class of Mama Chebet's drum (A), cover (B) and bucket (C) at the Kericho weighing station, and asks, 'If we can calculate their capacities, can we build our own and prove the maths works?' Groups of four design and build three models from cereal-box cardboard and old packaging: a cone, a

frustum of a cone and a frustum of a pyramid. They measure dimensions, calculate theoretical volume, then fill each model with water (lined with a plastic bag) or dry sand and measure the real capacity with a measuring jug. Groups calculate percentage error and discuss why measured values differ by about 5-10%, for example card thickness, gaps at the seams, imperfect shapes and spilled water. They take photos and video clips for documentation, add findings to the Driving Question Board and update the class model, which prepares them for the culmination in Lesson 8.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Visually Comparing the Capacity and Volume of Two Containers (Metric Units) <i>Related topic: not an exact match for this lesson.</i> Source: CK-12  Search ARES for similar videos  EXERCISE: Apply volume of solids Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Learners sit in their groups of four. Ms. Wanjiru shows a reminder card with the three Kericho containers (all 60 cm tall, 42 cm opening) and a sample half-finished cardboard frustum. Learners individually write a prediction in their notebooks: 'If I calculate the volume of my model correctly, will the water or sand I pour in match the answer exactly, a little less, or a little more? Why?' They then draw a quick sketch of the three models their group intends to make (cone, frustum of a cone, frustum of a pyramid) and label the dimensions they plan to use, choosing sizes small enough to measure easily (for example heights of 10 to 20 cm). Groups share their predictions with a partner group.	Say: 'Mama Chebet trusts the numbers for her drum, cover and bucket. Today you are going to test whether she is right to trust them.' Ask: 'When you pour water into a cardboard model, do you think the answer will match your calculation exactly? Write your prediction and one reason.' WAIT TIME: 15 seconds of silent thinking before anyone writes. Then cold-call 4 learners from different groups to share their predictions, without telling them whether they are right. Record 'exact', 'a bit less' and 'a bit more' as tallies on the board. Ask 2 more learners: 'What might make the real value differ from the calculated one?' and list their ideas on the board to return to in the Explain Phase. Remind learners about safe use of scissors.	Prediction with Reasoning (Think-Write-Pair-Share): committing to a written prediction and a reason before testing exposes learners' ideas about whether mathematical models match real objects. The tallies on the board give the class a shared prediction to compare with the evidence later, and the list of possible causes of difference gives learners hypotheses to check during the investigation.	Check the predictions for: (1) whether learners give a reason, not just a guess; (2) whether any mention of card thickness, gaps, spillage or imperfect shapes appears; (3) whether sketches show sensible dimensions labelled with units. Use a checklist to note learners who think a calculation must always match the real object exactly, and who should be asked to revisit their prediction during the Observe Phase. Collect the tallies to compare later.
Observe Phase  VIDEO: Visually Comparing the Capacity and Volume of Three Containers (Metric Units) <i>Related topic: not an exact match for this lesson.</i> Source: CK-12  Search ARES for similar videos	Groups build and test their models in four steps. (1) Design and construct: each group draws nets on cereal-box cardboard and builds a cone, a frustum of a cone (both radii and height chosen by the group) and a frustum of a pyramid with square faces. They tape the seams and line the inside with a plastic bag or tape so water does not leak. (2) Measure: they record R, r, h (perpendicular height, never the slant height) and	Circulate with a 'look-for' list. Prompt: 'Show me where on your model you are measuring h. Is it straight up and down, or along the slant?' Ask groups: 'How did you decide on these dimensions? What makes your calculation easier?' Stop the class twice: after construction, say 'Hold up your model and tell your partner which formula you will use and why.' WAIT TIME: 10 seconds before you take any response. Cold-call 3	Collaborative Investigation with Data Table (Planning and Carrying Out Investigations): learners generate their own data (dimensions, calculated volume and measured volume) and record it in a shared table structure. Because every group's numbers differ, learners cannot copy an answer; they must rely on the formula and the evidence. Rotating roles makes sure every learner contributes to measuring,	Observable indicators: (1) learners measure vertical height h, not slant height; (2) the correct formula is chosen for each solid; (3) the substitution uses $\pi = 22/7$ and squares the radii; (4) cm^3 is converted correctly to mL ($1 \text{ cm}^3 = 1 \text{ mL}$); (5) water is levelled at the brim and the jug is read at eye level; (6) the data table is complete with units. Note on a checklist which groups need help with the frustum of a pyramid formula, and

 EXERCISE: Apply volume of solids Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	the square sides, using rulers, and enter them in a data table. (3) Calculate: using $\pi = 22/7$ they work out the theoretical volume for each model with $V = 1/3 \pi r^2 h$, $V = 1/3 \pi h (R^2 + Rr + r^2)$ and $V = 1/3 h (A_1 + A_2 + \sqrt{A_1 A_2})$, then convert to mL. (4) Test: they fill each model to the brim with water or dry sand over a tray, pour the contents into a measuring jug and record the measured volume. One member photographs or videos each step for documentation, and groups record any challenges in measuring.	groups to state their formula and dimensions. During testing, remind learners to level the water at the brim and to read the jug at eye level. Ask: 'If your jug reads 1 mL less than expected, where could that mL have gone?' Make sure each learner has a role (builder, measurer, calculator, recorder/photographer) and rotates roles between models. Support groups who struggle to make a frustum of a pyramid by showing four trapezium faces on a net, but do not provide the dimensions. Reference Kericho figures only to compare shape: 'Does your frustum of a cone look like Mama Chebet's bucket? Which is the wide end?'	calculating and testing.	use mini-conferences of 2 minutes with those groups.
Explain Phase  VIDEO: Comparing the Capacity and Volume of Three Containers (Metric Units) <i>Related topic: not an exact match for this lesson.</i> Source: CK-12  Search ARES for similar videos  HTML: Volume of the Cone (PLIX) <i>Related topic: not an exact match for this lesson.</i> Source: CK-12  Search ARES for similar readings	Each group computes percentage error for each model: $(\text{measured} - \text{calculated} / \text{calculated}) \times 100$. For example, if a model has a calculated volume of 400 cm³ and a measured volume of 372 mL, the error is $28/400 \times 100 = 7\%$. Groups write a short explanation of the main sources of error in their own models (card thickness reducing internal size, gaps or bulges at seams, uneven bases, water or sand spilled or left sticking to the sides, reading the jug incorrectly). They then join a gallery walk in which each group posts its data table on the wall, and learners compare errors across groups. Finally, the whole class discusses: 'Do the formulas work for real objects? Why is the real answer a little different from the calculated answer?'	Model one percentage error calculation on the board using invented numbers, not a group's data, then have groups do their own. Say: 'Do not just tell me the error. Tell me which direction it went and what physical thing caused it.' Ask: 'Was your measured volume more or less than your calculated volume? Why would card thickness make it less?' WAIT TIME: 15 seconds. Cold-call 5 learners across groups to give one source of error each, and press: 'How would you change your build to reduce that error?' Pose the key challenge: 'Our errors are mostly within 5 to 10 percent. What does that tell you about whether the formulas are trustworthy?' Then link back to the phenomenon: 'Mama Chebet's drum, cover and bucket were built from the same shapes. Which of our models is most like her bucket, and what do our results suggest about her 58.52 L figure?' Discuss	Argument from Evidence with Error Analysis (Gallery Walk): learners treat percentage error as evidence about how well a model fits reality. Comparing data across groups shows patterns (for example, measured values are usually a little lower than calculated values because of card thickness), which helps learners reason about causes instead of seeing error as a mistake. They justify claims using their own data.	Look for: (1) correct percentage error calculation with the calculated volume as the denominator; (2) a stated cause of error linked to a physical feature of the model; (3) a claim such as 'the formula works because our error is small' supported by data; (4) willingness to question a result that is much larger than 10%. Use a 3-point rubric (calculates correctly / explains cause / uses evidence to judge the formula) and record scores for each group's explanation.

		any errors above 10 percent, asking the group to look for the likely cause, such as mistaking slant height for h or leaking seams, and invite them to retest.		
Driving Question Board (DQB) Creation  VIDEO: Volumes of cones intuition <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Volume of Cones (Flexbook) <i>Related topic: not an exact match for this lesson.</i> Source: CK-12  Search ARES for similar readings	Learners return to the Driving Question Board posted on the classroom wall. Each group writes at least one sticky note: either a new question that arose from the project (for example, 'How could we design a frustum that holds exactly 1 L?') or an answer card showing evidence from their model, such as 'Our cone model had a calculated volume of ____ cm³ and a measured volume of ____ mL, an error of ____%.' Learners move existing questions, such as 'Why does the drum hold three times as much as the cover?' and 'Why is the bucket not as big as the drum?', into the 'Answered', 'Partly answered' or 'Still to explore' columns, and attach evidence from their projects.	Say: 'Look at the DQB. Which of our questions did your models help to answer? Come and place your note beside it.' Cold-call 3 learners to read a question and justify moving it. WAIT TIME: 10 seconds after each question. Ask: 'What did building a frustum teach you that calculating alone did not?' Ask Mr. Kiprop's question again: 'Why is the bucket not as big as the drum?' and ask 2 learners to answer using their own frustum-of-a-cone model compared with a cylinder of equal height and top radius. Point out any new question about design, such as choosing dimensions for a required capacity, and tell learners that Lesson 8 will return to it.	Question Tracking on the Driving Question Board: moving questions between columns and attaching project evidence makes visible how understanding has grown from the first puzzle (why similar-sized containers hold different amounts) to being able to predict capacity. New questions keep inquiry going and show learners that science and mathematics build knowledge by testing ideas.	Observe whether sticky notes include evidence (numbers from models) rather than opinions, whether questions move to the correct column with a reason, and whether new questions are testable or design-oriented. Keep a tally of the number of learners who contribute a note and follow up with those who have not.
Model Building Phase  VIDEO: Visually Comparing the Capacity and Volume of Two Containers (Metric Units) <i>Related topic: not an exact match for this lesson.</i> Source: CK-12  Search ARES for similar videos  EXERCISE:	The class revises its single cumulative diagram, which began with the drum, cover and bucket. Using their projects, learners add: (1) a labelled sketch of each of their models next to the Kericho container it resembles (cone ~ cover B, frustum of a cone ~ bucket C); (2) the formulas used, with the dimensions measured; (3) a short note 'Calculated vs measured: error about 5-10%, caused by card thickness, seams, spills'; and (4) a design rule such as 'To get a required capacity, choose R, r and h, then check with $V = \frac{1}{3} \pi h (R^2 + Rr + r^2)$.' Each	Say: 'Our class model began as three containers. What new piece of evidence can we add today?' Invite one volunteer from each group, 4 groups in total, to add their annotation to the class diagram. WAIT TIME: 10 seconds before each volunteer speaks. Ask: 'Does the model now show how a cone, a frustum and a cylinder are connected?' Challenge: 'If Mama Chebet wanted a bucket with a bigger capacity but the same top opening, what could she change? Use your formula.' Cold-call 2 learners to answer. Check that every portfolio	Cumulative Model Revision: adding measured evidence and a design rule to the class diagram turns separate lessons into one explanation. Showing that formulas predict real capacities within a small, explainable error increases confidence in the model and prepares learners to use it to design containers.	Check that each portfolio contains: (1) labelled sketches of the three models; (2) the correct formula and dimensions for each; (3) calculated and measured volumes with the percentage error; (4) at least two named causes of error; (5) photos or video list. Use a rating scale (Exceeds / Meets / Approaches / Below Expectations) matched to the KICD rubric indicator 'ability to determine the volume and capacity of cones, pyramids, frustum of a cone and frustum of a pyramid'.

Volume of Cones Practice <i>Related topic: not an exact match for this lesson.</i> Source: CK-12 Search ARES for similar readings	group copies the update into a portfolio page together with photographs, a short video-clip list and a note of the challenges met during measurement. Groups agree where the models will be stored so they can be reused as learning resources.	has photos or clip lists, the data table and a one-sentence conclusion. Tell learners to bring all models and portfolios to Lesson 8, and thank them for taking care of the materials they collected.		
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D. TEACHER REFLECTION

After the lesson, consider: Which groups measured slant height instead of vertical height, and how did they notice? Were percentage errors mostly within 5-10%, and what did unusually large errors reveal about construction or measurement? Did every learner contribute to building, calculating and testing, or did some dominate? Did learners see percentage error as evidence about the model instead of as a failure? What could be changed (for example, better lining, sand instead of water, fixed dimensions) to improve accuracy next time? Which learners can now explain, in their own words, why the bucket holds less than the drum but more than the cover, and who needs more support before Lesson 8?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	What we observed: our cone, frustum of a cone and frustum of a pyramid models held measured amounts of water or sand that were close to, but usually a little different from, the volumes we calculated (record your own numbers: calculated ____ cm ³ , measured ____ mL, error ____%).
What did I learn?	What we learned: the formulas $V = \frac{1}{3} \pi r^2 h$, $V = \frac{1}{3} \pi h (R^2 + Rr + r^2)$ and $V = \frac{1}{3} h (A1 + A2 + \sqrt{A1 A2})$ predict the real capacity of containers within about 5-10%, and percentage error helps us judge how good the model is and why it differs.
How does this explain the phenomenon?	What we can now explain: the Kericho containers differ in capacity because their shapes differ, and mathematics lets us calculate and design a container of the capacity we need. Differences between calculated and measured values come from things like card thickness, imperfect seams and shapes, and spilled water or sand.

LESSON 8 (3 x 40-minute lessons (120 minutes, double lesson plus single lesson)): Culmination and Assessment: Explaining the Kericho Containers

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	To synthesise Lessons 1-7 into one annotated diagram of the volume relationships among cylinder, cone, pyramid and frustum, to give a final explanation of why the Kericho drum (A), cover (B) and bucket (C) hold such different amounts, and to assess all five specific learning outcomes.
Knowledge	By the end of the sub strand, the learner should be able to: a) determine the volume and capacity of cones and pyramids, b) compute volume and capacity of a frustum of a cone, c) calculate the volume and capacity of a frustum of a pyramid, d) apply volume and capacity in real-life situations, e) appreciate volumes and capacities in real life situations. Key facts: cylinder $V = \pi r^2 h$; cone $V = \frac{1}{3} \pi r^2 h$; pyramid $V = \frac{1}{3} \times \text{base area} \times \text{height}$; frustum of a cone $V = \frac{1}{3} \pi h (R^2 + Rr + r^2)$; frustum of a pyramid $V = \frac{1}{3} h (A_1 + A_2 + \sqrt{A_1 A_2})$. With $\pi = 22/7$: drum A = 83.16 L, cover B = 27.72 L, bucket C = 58.52 L; the drum holds exactly 3 times the cover; bucket = large cone 83,160 cm ³ minus small cone 24,640 cm ³ .
Skills	NGSS-style practices embedded: Developing and Using Models (build the final annotated diagram), Constructing Explanations (write the final explanation using evidence), Engaging in Argument from Evidence (defend the choice of formula), Analyzing and Interpreting Data (compare Lesson 1 predictions with actual values and project results), Using Mathematics and Computational Thinking (solve the assessment items), Communicating (group project presentations).
Attitudes	Values: Unity - the learner displays tolerance of different opinions while working with peers on the final diagram. Respect - the learner listens to others' opinions while discussing formulas and sources of error in project presentations. Appreciation of the usefulness of volume and capacity in tea farming, storage and cooking. Core competencies: Critical Thinking and Problem Solving; Creativity and Imagination; Communication and Collaboration. PCI: Environment and technology - learners use containers and materials from the local environment responsibly.
Key Inquiry Question	Why is the knowledge of volume and capacity useful?
Purpose in Storyline	This is the culmination of the storyline. Learners return to Mama Chebet's drum (A), cover (B) and bucket (C) in Kericho and bring together everything gathered in Lessons 1-7: the pouring evidence for one third, the pyramid, the large-minus-small method for frustums, the problem solving and the project models. They complete the DQB, compare their Lesson 1 predictions with the real values, finish the cumulative model as one annotated diagram, and show what they can now do independently in a written class activity.
Safety Notes	Keep floors dry; wipe up any spilled water from the project verification immediately to prevent slipping. Learners handle the model containers, scissors and rulers carefully. Tidy and reuse card, sand and containers from the project. Carry water containers in pairs. Nobody is to taste any water or materials used in the models.

B. LESSON OVERVIEW

Ms. Wanjiru opens by reminding learners of Mr. Kiprop's question at the Kericho weighing station: why is the bucket not as big as the drum when they look the same size? Over three 40-minute periods, learners (1) present their Lesson 7 project models, with calculated versus measured capacities and explanations of error, (2) revisit their Lesson 1 predictions and the DQB, and build the final class diagram that shows the cylinder (83.16 L) as exactly 3 x the cone (27.72 L), the bucket (58.52 L) as large cone (83,160 cm³) minus small cone (24,640 cm³), and the pyramid and its frustum following the same one-third and large-minus-small ideas, (3) write their Final Explanation of the phenomenon, and (4) complete a written class activity covering all five specific learning outcomes. This lesson does not teach new content; it

consolidates and assesses.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Volumes of cones intuition <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Apply volume of solids Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Learners retrieve their Lesson 1 prediction sheets (which of the drum A, cover B and bucket C holds most, and their estimates). Individually they silently write a new prediction in 3 minutes: 'Without calculating, rank A, B and C and write the ratio of the cover to the drum. Then predict: if Mama Chebet wanted a new frustum bucket with a top radius of 21 cm, a bottom radius of 14 cm, but only 30 cm tall, would the capacity be half of 58.52 L, more, or less? Why?' They then place their Lesson 1 and new predictions side by side and mark which ideas they now want to change. Next, in groups of 4, learners also predict which part of today's work they expect to be hardest and which formula they will need first.	Hand back the Lesson 1 prediction sheets without comment. Say: 'Mr. Kiprop still asks why his bucket is not as big as the drum. Before we answer him, write what you think now, and mark in a different colour anything you would change from Lesson 1.' WAIT TIME 15 seconds after posing the frustum-height question before anyone writes. Walk the room and note, without correcting, the reasoning used. Cold-call 4 learners, choosing at least 2 with different predictions, to give their ranking and one reason. Ask: 'Who changed their mind since Lesson 1? What made you change it?' Cold-call 3 more for the answer. Record all predictions on the board in a tally without saying which is right. Do not reveal answers: the aim is to expose reasoning that the evidence from earlier lessons can now test.	Prediction comparison (then-and-now). Learners compare Lesson 1 thinking with current thinking, which makes the growth in their reasoning visible, and shows that predictions based on 'looks the same size' (same height and opening) were challenged by evidence about shape. It also surfaces any lingering misconception, for example that equal height and opening mean equal capacity, or that halving the height halves the frustum volume, for the later phases to test.	Observable indicators: (1) learners annotate their Lesson 1 sheet with at least one change and a reason; (2) at least 3 of the 4 cold-called learners rank cover < bucket < drum and mention shape or one third, not just size; (3) note learners who still say 'they are the same size so they hold the same' and plan to pair them with a peer during the diagram building; (4) record how many predict that the 30 cm bucket holds half as much, to address at the Explain phase.
Observe Phase  VIDEO: Visually Comparing the Capacity and Volume of Two Containers (Metric Units) <i>Related topic: not an exact match for this lesson.</i> Source: CK-12  Search ARES for similar videos	Learners observe evidence in two parts. Part 1 (about 40 minutes): project presentations. Each group (a 4-minute presentation plus 1-minute questions) shows its model container from Lesson 7 (cone, pyramid or frustum), gives its measurements, the formula used, the calculated volume and capacity, the measured volume from filling with water or sand, the percentage error, and 2 causes of error such as card thickness, imperfect shape, spilled water or sand. The audience records each	Set a timer visible to all, and say: 'Presenters: formula first, then numbers, then what went wrong and why. Audience: you are scientists checking their evidence; each of you must write one question.' After each presentation WAIT TIME 10 seconds before taking questions, and cold-call 1 learner from the audience for a question and 1 for an error cause they agree with. Ask each group: 'Was your measured volume higher or lower than calculated? What does that tell you about	Data table and peer review of evidence (scientific discourse). Learners collect all group results in one class table and critique the evidence, which builds the understanding that models give theoretical values and measurement gives real values, and that percentage error shows the quality of agreement. Re-computing the Kericho values from the same data lets learners trace the explanation of the phenomenon back to a calculation rather than to memory.	Observable indicators: (1) each presenter states the correct formula for the shape and substitutes values with $\pi = 22/7$; (2) calculated volume is converted correctly to litres ($1000 \text{ cm}^3 = 1 \text{ L}$); (3) percentage error is computed correctly as $(\text{difference} / \text{calculated}) \times 100$ and lies within roughly 5-10%, or learners give a reason when it does not; (4) learners give at least 2 plausible error causes; (5) in Part 2, at least 90% of learners get 83.16 L, 27.72 L and 58.52 L and the ratio 3:1

 EXERCISE: Volume of cylinders, spheres, and cones word problems Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	group's data in a class table: model, calculated capacity, measured capacity, percentage error. Part 2 (about 15 minutes): the class re-observes the original evidence. Learners are given the three Kericho containers' data card (A: $r = 21$ cm, $h = 60$ cm; B: $r = 21$ cm, $h = 60$ cm; C: $R = 21$ cm, $r = 14$ cm, $h = 60$ cm) and recalculate: drum A 83.16 L, cover B 27.72 L, bucket C 58.52 L. They find $83.16 / 27.72 = 3$ and $58.52 / 83.16$ which is about 0.70, then complete a data-table row comparing their Lesson 1 predicted values with these actual values.	where the error came from?' Ensure the expected error range of about 5-10% is discussed, and push groups with a larger error to propose a reason and a fix. For Part 2, move among groups asking: 'Which step in your drum calculation used r squared? Show me where 21×21 appears.' Cold-call 3 learners to state the cylinder-to-cone ratio and 3 to state the bucket as a fraction of the drum. Do not reteach the formulas; ask learners to point to the page or the model from the earlier lesson that supports each value.		without help. Note groups needing support for the targeted follow-up in the assessment.
Explain Phase  VIDEO: Visually Comparing the Capacity and Volume of Three Containers (Metric Units) <i>Related topic: not an exact match for this lesson.</i> Source: CK-12  Search ARES for similar videos  EXERCISE: Apply volume of solids Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Learners work in groups of 4 to write the Final Explanation of the phenomenon on a Claim-Evidence-Reasoning organiser. Claim: why the drum A, cover B and bucket C look nearly the same but hold different amounts. Evidence: the data (same $h = 60$ cm and opening radius 21 cm; 83.16 L, 27.72 L, 58.52 L; 3:1 ratio; pouring evidence from Lesson 2; the project class table). Reasoning: shape decides volume; the cone is one third of the cylinder with the same r and h, $V = \frac{1}{3} \pi r^2 h$; the bucket is a frustum, which equals the large cone ($r = 21$, $h = 180$, 83,160 cm³) minus the small cone ($r = 14$, $h = 120$, 24,640 cm³) = 58,520 cm³ = 58.52 L; and the same ideas of one third and large-minus-small hold for the pyramid ($\frac{1}{3} \times$ base area $\times$ height) and the frustum of a pyramid. Groups then answer Mr. Kiprop in 2-3 sentences: 'Why is my bucket not as big as the drum?' and use the explanation to say how to design a frustum container	Say: 'Imagine Mr. Kiprop is standing here. Your explanation must convince him using numbers, not just words.' Display the CER headings. WAIT TIME 15 seconds after posing: 'Which single idea explains why the cover holds one third of the drum?' before taking responses. Circulate and use probes: 'Where does the 180 cm come from? Show me the similar triangles or the 2:3 ratio.' 'Why can't you just multiply the drum by 0.7 to get the bucket in general?' 'What would change if the bottom radius were 10 cm instead of 14 cm?' Cold-call 4 groups to read their reasoning aloud, and ask 2 other groups to say whether the reasoning connects evidence to claim. Press for precision: volume versus capacity, h as the vertical height, never the slant height. Point out the one-third thread connecting cone, pyramid and the large-minus-small method for both kinds of frustum, without re-deriving anything.	Claim-Evidence-Reasoning (CER) with Mr. Kiprop as the authentic audience. CER forces learners to link data to the mathematical rule that explains it, so the Final Explanation rests on evidence collected across Lessons 1-7. Applying the explanation to a new design question (a 30 cm bucket, a required capacity) shows that they can now predict as well as explain, answering the driving question.	Observable indicators: (1) the claim states that shape, not height or opening alone, determines volume; (2) evidence cites at least 3 numbers from the Kericho data (83.16 L, 27.72 L, 58.52 L, or the cone volumes); (3) reasoning mentions $V = \frac{1}{3} \pi r^2 h$ and the large-minus-small cone method; (4) the learner correctly gets 29.26 L for the 30 cm bucket and explains the effect of height; (5) correct use of units (cm³ to L). Use a 3-level checklist (complete, partial, not yet) for each of the 5 items and follow up with the groups marked 'not yet' during the diagram building.

	of a required capacity, for example choosing R, r and h and testing the formula. Groups finish by revisiting the 30 cm bucket prediction and test it: $V = \frac{1}{3} \times (22/7) \times 30 \times 931 = 29,260 \text{ cm}^3 = 29.26 \text{ L}$, which is half of 58.52 L, and discuss that this works only because R and r stay the same while the height halves.			
Driving Question Board (DQB) Creation  VIDEO: Comparing the Capacity and Volume of Three or More Containers (Metric Units) <i>Related topic: not an exact match for this lesson.</i> Source: CK-12  Search ARES for similar videos  EXERCISE: Volume of cylinders, spheres, and cones word problems Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Learners gather at the DQB posted on the classroom wall. Each learner takes the sticky-note questions the class posted since Lesson 1 (for example: 'Why does the drum hold more than the cover?', 'Does the bucket hold two thirds of the drum?', 'Can we design a frustum that holds exactly a given capacity?') and, in groups, sorts them into three columns: Answered (with the lesson and evidence that answered it), Partly answered, and New questions. Each learner writes an answer card for at least one question, citing evidence such as 'cover = $27.72 \text{ L} = \frac{1}{3}$ of 83.16 L'. Groups also add 1 new question for further study, for example: 'How would the capacity change for a tea-leaf basket with an oval opening?' or 'How much maize would the bucket hold at 0.75 kg per litre?' Learners then complete a 'Then and Now' summary: what I predicted in Lesson 1, what I now know, and what I can explain to others.	Say: 'This board is our record of how we learned. Move your questions to the column they belong to, and show me the evidence.' WAIT TIME 10 seconds after asking: 'Which question took the most lessons to answer, and why?' Cold-call 5 learners to defend where they placed a question and ask a different learner each time: 'Do you agree? What evidence supports that?' For questions left partly answered, ask: 'What data or model would we need to answer it?' Praise precise use of evidence, not just correct answers. Note any unanswered questions as honest next steps and ask 2 learners to connect one to the driving question: 'How can we use mathematics to design a frustum-shaped container that holds exactly the capacity we need?' Photograph the completed DQB for the portfolio.	DQB sorting and revisiting (tracking growing understanding). Moving questions from open to answered with evidence shows learners how their understanding developed over the unit, and new questions highlight that knowledge in volume and capacity leads to further inquiry. It is a metacognitive record of change from Lesson 1.	Observable indicators: (1) every learner has at least 1 answer card citing specific evidence (a number or a lesson activity); (2) learners correctly place questions as answered or partly answered, with justification; (3) at least 1 new question per group that is testable using volume or capacity ideas; (4) the 'Then and Now' summary shows a concrete change, for example 'I thought the bucket was nearly as big as the drum; now I know it is about 70% of it'; (5) learners can explain why volume and capacity knowledge is useful in tea collection, storage or cooking, linking to the key inquiry question.
Model Building Phase  VIDEO: Volume of pyramids intuition <i>Related topic: not</i>	Part 1 (about 20 minutes): Learners build the final cumulative class diagram on a large chart. Each group is allocated a section and draws and annotates it: (1) cylinder (drum A, $r = 21$, $h = 60$, 83.16 L) beside the cone (cover B,	Say: 'This is our final model. Every shape must show its volume, its formula, and how it connects to the others. If an arrow has no reason, add the reason.' WAIT TIME 15 seconds after: 'How is the pyramid frustum like the cone frustum?'	Cumulative model revision and comparison (L1 versus final). The final annotated diagram unites all shapes through two linked ideas: one third and large-minus-small, so learners see the structure of the content rather than five unrelated	Observable indicators: (1) the diagram shows the correct values (83.16 L, 27.72 L, 58.52 L, $83,160 \text{ cm}^3$ minus $24,640 \text{ cm}^3$, 12 L, 84 L) and both linking ideas with a reason for each arrow; (2) the written activity is marked against

<i>an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum) Search ARES for similar videos  HTML: Volume of the Cone (PLIX) <i>Related topic: not an exact match for this lesson.</i> Source: CK-12 Search ARES for similar readings	same r and h, 27.72 L) with the label 'cone = $\frac{1}{3}$ cylinder, 3 cones fill the drum'; (2) the bucket C drawn as the large cone ($r = 21$, $h = 180$, 83,160 cm³) with the small cone ($r = 14$, $h = 120$, 24,640 cm³) removed, labelled 58,520 cm³ = 58.52 L, with the 2:3 radius ratio; (3) the cuboid and pyramid (30 x 30 x 40: 36,000 cm³ and 12,000 cm³ = 12 L); (4) the maize bin frustum of a pyramid ($A_1 = 3,600$, $A_2 = 900$, $h = 40$, 84 L, 63 kg of maize), shown as large pyramid 96,000 cm³ minus small pyramid 12,000 cm³; (5) a formula strip with all five formulas and a unit conversion strip (1 cm³ = 1 mL, 1000 cm³ = 1 L, 1 m³ = 1000 L). Learners add arrows linking the 'one third' idea and the 'large minus small' idea across shapes. Part 2 (about 25 minutes): written class activity done individually in silence covering all five specific learning outcomes. Items: (a) a cone of $r = 7$ cm and $h = 12$ cm: find the volume and capacity in mL (616 cm³ = 616 mL); (b) a square pyramid with base side 18 cm and height 25 cm: volume and capacity (2,700 cm³ = 2.7 L); (c) a bucket (frustum of a cone) with $R = 15$ cm, $r = 10$ cm and $h = 28$ cm: capacity ($\frac{1}{3} \times \frac{22}{7} \times 28 \times 475 = 13,933.33$ cm³, about 13.93 L); (d) a frustum of a square pyramid with top side 40 cm, bottom side 20 cm and $h = 30$ cm: volume and capacity ($\frac{1}{3} \times 30 \times (1,600 + 400 + 800) = 28,000$ cm³ = 28 L) and the mass of maize that fills it at 0.75 kg/L (21 kg); (e) a real-life item: how many cups of 462 cm³ can the Kericho drum A (83,160 cm³) fill, and a short written statement about why volume and capacity knowledge is useful to a Kericho	before cold-calling 3 learners. Compare the chart with the Lesson 1 model, which had only the three containers and predictions, and ask 2 learners to describe three differences, then ask: 'What does this final model let you do that the first one could not?' Check that h is the vertical height, units are consistent, and $\pi = \frac{22}{7}$ throughout. For the written activity, say: 'Work alone. Show every step: formula, substitution, answer with units.' No talking; circulate silently and note common errors without correcting. Provide read-aloud support or extra time on request. After collecting scripts, return to the group chart and cold-call 4 learners to say what they will remember about why volume and capacity matter. Close by linking to the key inquiry: 'Why is the knowledge of volume and capacity useful?', and have 3 learners answer with a Kenyan example.	formulas. Comparing it with the Lesson 1 model shows how a visual guess became an evidence-based explanatory and predictive model, and the individual written activity checks whether each learner can use the model on their own.	the five specific learning outcomes, one for each item: (a) cone, (b) pyramid, (c) frustum of a cone, (d) frustum of a pyramid, (e) application and appreciation; (3) scoring per outcome using the KICD rubric levels: Exceeds (correct, with explanation and varied reasoning), Meets (correct formula, substitution and units), Approaches (minor errors, such as unit conversion or using slant height), Below (major errors or missing steps); (4) at least 80% of learners reach Meets on 4 of the 5 items; (5) the exit self-rating matches the evidence in their scripts, and learners at Approaches or Below are scheduled for targeted revision on the specific formula.
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	tea farmer or a cook in Nairobi. Part 3 (about 10 minutes): learners complete an exit self-rating on each of the five learning outcomes (can do alone, can do with help, not yet).			
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D. TEACHER REFLECTION

After the lesson, review the class table, written scripts and DQB: (1) Which outcomes did most learners meet, and which items (cone, pyramid, frustum of a cone, frustum of a pyramid, application) caused the most errors? Were the errors in choosing the formula, in substituting r, R and h (for example using slant height), or in unit conversion (cm3 to L)? (2) Did learners' Final Explanation use evidence and reasoning, or only restate the facts? Which groups still said that equal height and opening mean equal capacity? (3) How did the percentage errors in the project presentations compare with the expected 5-10%, and did learners give sound causes? (4) Did the comparison between Lesson 1 predictions and the actual values (83.16 L, 27.72 L, 58.52 L) show real conceptual change? Which learners need further support, and what will you do next, such as a short revision session or mixed-ability pairs? (5) What would you change in the sequence of Lessons 1-8 next time, and which Kenyan contexts motivated learners most? Record 2-3 notes for next year's planning.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	We observed that the Kericho drum (A), conical cover (B) and metal bucket (C) all have a height of about 60 cm and an opening radius of 21 cm, yet hold 83.16 L, 27.72 L and 58.52 L respectively. We also observed the project models, whose measured capacities came within about 5-10% of the calculated ones.
What did I learn?	We learned that shape decides volume: the cone is exactly one third of the cylinder, $V = \frac{1}{3} \pi r^2 h$, and the pyramid is one third of its matching cuboid. A frustum is a large cone or pyramid with a smaller one removed, which gives $V = \frac{1}{3} \pi h (R^2 + Rr + r^2)$ and $V = \frac{1}{3} h (A_1 + A_2 + \sqrt{A_1 A_2})$. We learned how to convert cm3 to litres and how to calculate percentage error.
How does this explain the phenomenon?	We can explain that the three containers look alike because they share the same height and opening, but the drum holds exactly 3 times the cover (83.16 L against 27.72 L). The bucket holds 58.52 L, between them, because it is the large cone (83,160 cm3) minus the small cone (24,640 cm3). Using the same reasoning for pyramids and frustums, we can design a frustum container that holds exactly the capacity needed, which is why volume and capacity knowledge is useful to Kericho tea farmers and many others.

DIFFERENTIATION AND INCLUSION		
Learner Need	Support Strategies	Extension Strategies
English Language Learners	Provide bilingual glossaries; allow use of first language for initial model drawing.	Write extended explanations of observations; lead group discussions in English.
Students with learning difficulties	Provide structured note frames; pair with a supportive partner during investigations.	Mentor peers; create additional visual models to explain concepts.
Gifted and advanced learners	Access to additional reading materials; challenge questions during DQB.	Lead model-building discussions; research and present extension topics to class.